

# **From Source to Execution: Design and Implementation of a Statically-Typed Programming Language with Type Inference**

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## ABSTRACT

This report presents the design and implementation of a small, statically-typed interpreted programming language realised through a four-stage compiler pipeline. The language provides four primitive types — Integer, Float, Boolean, and String — and supports arithmetic and equality expressions, variable assignment with type inference from first use, conditional and iterative control flow, user-defined functions with value parameter passing, and a built-in callable `print()` function. Types are inferred from program context rather than declared explicitly, combining static type safety with a declaration-free syntax. The pipeline consists of a hand-written lexer, a recursive-descent parser that constructs an abstract syntax tree together with an inspectable parse-tree view, a static type checker that performs inference and rejects ill-typed programs before execution, and a tree-walking interpreter. Both a command-line runner and an interactive desktop interface expose each pipeline stage for inspection. The report documents the formal grammar, the type inference algorithm, the system architecture, and the automated test suite of 75 tests that validates the complete implementation.

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# CHAPTER 1

## INTRODUCTION

Programming languages define the formal contract between a programmer’s intent and machine execution. Building a language processor from source text through to evaluated output requires a coherent pipeline of components, each of which must correctly transform its input before passing it downstream. Lexical analysis identifies atomic tokens; parsing imposes hierarchical structure; static type checking validates semantic correctness before execution; and interpretation evaluates the validated program. When these stages are connected cleanly, errors are caught at the earliest possible point, and the overall system remains tractable to reason about, test, and extend.

This report presents the design and implementation of a small statically-typed interpreted language that realises this pipeline in full. The language supports four primitive types — Integer, Float, Boolean, and String — and provides arithmetic and equality expressions, assignment, conditional and iterative control flow, user-defined functions with value parameter passing, and a built-in callable `print()` function. Types are inferred from program context rather than declared explicitly, combining the safety of static typing with the conciseness of a declaration-free syntax. The implementation spans three principal components developed in pipeline order: the lexer and symbol table, the recursive-descent parser together with parse-tree and abstract-syntax-tree views, and the type checker with interpreter. Each component was completed before the next was begun, since each stage consumes the structured output of its predecessor.

The implemented system can be invoked from the command line or through an interactive desktop interface, both of which surface the five pipeline stages — tokens, parse tree, AST, inferred type table, and execution output — for inspection. The parse tree is provided as a grammar-oriented view for submission and debugging, while the AST remains the executable intermediate representation consumed by the type checker and interpreter. An automated regression suite of 75 tests verifies correctness across all stages and language features.

The remainder of this report is organised as follows. Chapter 2 presents the language design, token specification, grammar, and operator precedence rules. Chapters 3 and 4 cover the front-end implementation in pipeline order: lexical analysis and the symbol table, then recursive-

descent parsing and the AST. Chapter 5 explains the type inference algorithm and the semantic rules used by the checker. Chapter 6 presents the system architecture and implementation, including the end-to-end pipeline, project layout, and technology stack. Chapter 7 discusses testing and verification, and Chapter 8 concludes the report.

## CHAPTER 2

### LANGUAGE DESIGN AND GRAMMAR

#### 2.1 Types

The language provides four primitive types, chosen to cover the core domains of numeric computation, logical control flow, and text manipulation while keeping the type system tractable and the semantics well-defined.

##### 2.1.1 *Integer*

The Integer type represents whole numbers such as 0, 7, and 42. Integer literals are written as one or more decimal digits without quotation marks or a decimal point.

##### 2.1.2 *Float*

The Float type represents real-valued numbers written with a decimal point, such as 3.14 or 0.5. Float values are used exclusively with the dot-suffixed arithmetic operators ( $+. , -. , *. , /. ,$ ), following the Caml Light convention. Integer division ( $/$ ) performs floor division and returns Integer: for example,  $10 / 3$  evaluates to 3 : Integer.

The language intentionally does not perform implicit numeric promotion between Integer and Float values. This design decision follows directly from the assignment specification, which requires “no overloading with the usual precedence of multiplication and division over addition and subtraction.” In a language where a single  $+$  operator accepts both  $(Integer, Integer) \rightarrow Integer$  and  $(Integer, Float) \rightarrow Float$ , the operator effectively carries multiple type signatures — the compiler must inspect both operand types to determine the result type. This constitutes a form of operator overloading, even when described as “numeric promotion” or “widening coercion.” To eliminate all such ambiguity, the language follows the Caml Light convention of using separate operator tokens for integer and floating-point arithmetic:  $+$ ,  $-$ ,  $*$ ,  $/$  operate exclusively on Integer operands, while  $+. , -. , *. , /. ,$  operate exclusively on Float operands. Mixed-type arithmetic such as  $1 + 2.5$  is rejected by the type checker. Under this design, every operator token uniquely determines its operand types and result type — the compiler never needs to inspect operand types to resolve operator meaning, which is the defining property of a

language without operator overloading.

### 2.1.3 *Boolean*

The Boolean type represents logical truth values. The two Boolean literals are `true` and `false`. Boolean values are used in conditions and as results of comparison expressions.

### 2.1.4 *String*

The String type represents sequences of characters enclosed in double quotes, for example `"hello"`. Strings can be assigned to variables and printed through the built-in `print()` function. The surrounding quotes are stripped by the parser when the `Literal` node is created, so the value stored internally is the bare character sequence without surrounding quotes. At runtime, the built-in `print()` function re-adds double quotes around string values, so `print("plc")` outputs `"plc" : String`.

### 2.1.5 *Void*

Although the language exposes exactly four programmer-visible types, the implementation introduces a fifth value, `Void`, within the `LanguageType` enumeration. Its sole purpose is to provide a well-defined return type for functions that contain no `return` statement, allowing the symbol table to record a consistent entry for every function regardless of whether it produces a value. `Void` cannot appear in source code and is never visible to the programmer; it is a bookkeeping device used internally by the type checker. Therefore, `Void` does not extend the programmer-visible type system required by the assignment.

## 2.2 Static Typing

The language employs static typing, meaning that type errors are detected at type-checking time rather than at runtime. No explicit variable type declarations are required. Instead, the type checker infers types from literals, assignments, operations, function calls, and return statements. Once a variable type is established, later assignments must remain consistent with that type. For example, the following program is rejected at the type-checking stage before any execution occurs:

```
1 x = 5;
2 x = "hello";    (* TypeError: Cannot assign String to variable 'x' of
   type Integer *)
```

This contrasts with dynamically-typed languages, where the same program would run without error until (or unless) a type-dependent operation was attempted.

## 2.3 Token Specification

Table 2.1 summarises the major token classes used by the language.

Table 2.1. Token Specification

| Token Category                 | Example Lexemes                            |
|--------------------------------|--------------------------------------------|
| INT_LITERAL                    | 0, 10, 42                                  |
| FLOAT_LITERAL                  | 3.14, 20.5                                 |
| BOOL_LITERAL                   | true, false                                |
| STRING_LITERAL                 | "hello"                                    |
| IDENTIFIER                     | variable and function names such as x, add |
| Arithmetic operators (integer) | +, -, *, /                                 |
| Float arithmetic operators     | +, -, *, /                                 |
| Comparison operators           | ==, !=                                     |
| Assignment operator            | =                                          |
| Keywords                       | if, else, while, def, return               |
| Delimiters                     | ( ) { } , ;                                |

## 2.4 Context-Free Grammar

The language grammar can be described by the context-free grammar  $G = (V_N, V_T, P, S)$ , where  $V_N$  is the set of non-terminals,  $V_T$  is the set of terminals,  $P$  is the set of production rules, and  $S$  is the start symbol program. The grammar is implemented through recursive-descent parsing functions.

### 2.4.1 Program Structure

```

1 program    -> statement* EOF
2 statement  -> assignment
3           | if_stmt
4           | while_stmt
5           | func_def
6           | return_stmt
7           | expr ';'

```

### 2.4.2 Expressions

```

1 expr       -> additive (( '=' | '!=' ) additive)?
2 additive   -> term (( '+' | '-' | '+.' | '-.' ) term)*
3 term       -> factor (( '*' | '/' | '*.' | '/.' ) factor)*
4 factor     -> '-' factor      (* integer unary negation *)

```

```

5 | ' - . , ' factor      (* float unary negation *)
6 | INT_LITERAL
7 | FLOAT_LITERAL
8 | STRING_LITERAL
9 | BOOL_LITERAL
10 | IDENTIFIER
11 | IDENTIFIER ' ( ' args? ' ) '
12 | ' ( ' expr ' ) '

```

The operators `+`, `-`, `*`, and `/` operate exclusively on Integer operands. The dot-suffixed operators `+. ,`, `-. ,`, `*. ,` and `/. ,` operate exclusively on Float operands. This follows the Caml Light convention of using distinct operator tokens for integer and float arithmetic, eliminating the need for implicit numeric promotion. Unary negation is similarly split: `-x` negates an Integer (desugared to `0 - x`), and `-.x` negates a Float (desugared to `0.0 -. x`).

**Integer division and division by zero.** `/` is floor (integer) division: both operands must be Integer, the result is Integer, and the value is  $\lfloor a/b \rfloor$ . For example, `7 / 2` evaluates to 3 : Integer. Float division uses the distinct operator `/. ,` and requires two Float operands. Division by zero (`x / 0` or `x /. 0.0`) is detected at runtime and raises a `RuntimeError`.

### 2.4.3 Boolean Expressions

Comparison operators `==` and `!=` are the lowest-precedence binary operators and produce Boolean values. Because they are part of the general `expr` rule, Boolean expressions may appear both as control-flow conditions and as ordinary expressions. Their inferred type is Boolean, allowing comparison results to be assigned to variables, passed as arguments, or printed:

```

1 flag = (x == y);      (* flag inferred as Boolean *)
2 print(a != b);        (* prints true/false : Boolean *)

```

**Strict arithmetic operands.** The specification states that `==` and `!=` operate between two arithmetic expressions. We interpret “arithmetic” strictly as Integer and Float, consistent with the no-overloading rule. Both operands must also share the same numeric type (no implicit coercion). Therefore `Integer == Integer` and `Float == Float` are well-typed; `Integer == Float`, `Boolean == Boolean`, and `String == String` are all rejected at type-check time.

### 2.4.4 Statements

```

1 assignment -> IDENTIFIER '=' expr ';'
2 if_stmt    -> 'if' '(' expr ')' block ('else' block)?
3 while_stmt -> 'while' '(' expr ')' block
4 return_stmt -> 'return' expr ';'

```

```
5 block      -> '{' statement* '}'
```

### ***Design note: `print` as a built-in callable***

`print` is implemented as a normal function-call expression rather than as a dedicated statement form. Lexically, `print` is tokenised as an `IDENTIFIER`; syntactically, `print(x)` is parsed through the same `FunctionCall` path used for user-defined calls. Semantically it is a unary built-in of type  $T \rightarrow ()$ : it accepts one argument of any programmer-visible type, emits value : Type immediately at runtime, and returns the internal `Void` type. As with any void-returning call, `print(...)` may be used as a standalone expression statement but not as the right-hand side of an assignment.

## **2.4.5 Function Definitions and Calls**

```
1 func_def -> 'def' IDENTIFIER '(' params? ')' block
2 params   -> IDENTIFIER (',' IDENTIFIER)*
3 args     -> expr (',' expr)*
```

Function calls use value parameter passing. This means the argument values are evaluated before the call and copied into the function's local environment.

The built-in `print()` function uses exactly the same surface syntax as other calls:

```
1 print(x);
2 print(3.14);
```

**Return type uniformity.** A function's return type is inferred from its return statements. All return expressions within a function body must yield the same type; mixed return types are a compile-time error. A function with no return statement is assigned the special internal type `Void` and may only be invoked as a standalone statement, not used as an expression. Assigning the result of a `Void` function to a variable (e.g. `x = f()`; where `f` returns nothing) is rejected by the type checker.

## **2.5 Operator Precedence and Associativity**

Operator precedence is encoded structurally in the recursive-descent parser. Comparison is lowest, addition and subtraction are next, and multiplication and division have the highest precedence among binary operators. Arithmetic operators are left-associative.

Table 2.2. Operator Precedence (1 = lowest, 3 = highest) and Associativity

| Precedence  | Operators                                                         | Associativity                                   |
|-------------|-------------------------------------------------------------------|-------------------------------------------------|
| 1 (lowest)  | <code>==</code> , <code>!=</code>                                 | non-associative (one comparison per expression) |
| 2           | <code>+</code> , <code>-</code> , <code>+</code> , <code>-</code> | left-associative                                |
| 3 (highest) | <code>*</code> , <code>/</code> , <code>*</code> , <code>/</code> | left-associative                                |

## 2.6 Design Decisions: No Overloading

The assignment specification states: “No overloading with the usual precedence of multiplication and division over addition and subtraction. Hence no explicit declaration.” This section explains how the language design satisfies this requirement and why alternatives were rejected.

**Definition of overloading.** An operator is overloaded when the same syntactic symbol carries more than one type signature, requiring the compiler to inspect operand types to determine the operator’s meaning or result type. Under this definition, a language where `+` means both integer addition  $(Integer, Integer) \rightarrow Integer$  and floating-point addition  $(Float, Float) \rightarrow Float$  — or worse, mixed-type addition  $(Integer, Float) \rightarrow Float$  — overloads the `+` symbol.

**Our approach: distinct operator tokens.** Following the Caml Light convention, this language assigns a unique token to each arithmetic meaning. Integer arithmetic uses `+`, `-`, `*`, `/`; floating-point arithmetic uses `+.` , `-.`, `*.`, `/.` Each token has exactly one type signature. The type of any expression can be determined from the operator token alone, without inspecting operand types. This is the strongest possible interpretation of “no overloading.”

**Rejected alternative: implicit numeric promotion.** An alternative design would use a single set of operators and silently promote Integer operands to Float when mixed with a Float operand (e.g.  $1 + 2.5 \rightarrow 3.5 : Float$ ). Proponents of this approach argue that promotion is not overloading because the operator retains a single “arithmetic addition” meaning. However, this argument is difficult to sustain formally: the operator `+` would have three distinct input-type combinations —  $(Integer, Integer)$ ,  $(Integer, Float)$ , and  $(Float, Float)$  — with two different result types, a dispatch table that the compiler must resolve from operand types, which is the hallmark of overloading. The assignment specification also states “hence no explicit declaration,” implying that type inference should be unambiguous; implicit promotion introduces cases where the inferred result type depends on operand types rather than on the operator itself.

**Connection to type inference.** Because every operator token uniquely determines its operand and result types, the type inference algorithm described in Chapter 5 can determine the type

of any expression without backtracking, unification, or constraint solving. This determinism directly satisfies the specification’s coupling of “no overloading” with “no explicit declaration.”

Table 2.3 contrasts the two designs.

Table 2.3. Comparison of operator design approaches

| Expression             | This language (separate to-kens)                       | Promotion-based language                                                                          |
|------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| $3 + 4$                | $7 : \text{Integer} (+ \rightarrow \text{Integer})$    | $7 : \text{Integer}$                                                                              |
| $3.0 +. 4.0$           | $7.0 : \text{Float} (+. \rightarrow \text{Float})$     | N/A (no $+. $ operator)                                                                           |
| $3 + 4.0$              | TypeError: $+$ requires two Integer operands           | $7.0 : \text{Float}$ (implicit promotion)                                                         |
| $10 / 3$               | $3 : \text{Integer}$ (floor division)                  | $3.333\dots : \text{Float}$ (or $3 : \text{Integer}$ , design-dependent)                          |
| Type signatures of $+$ | One: $(\text{Int}, \text{Int}) \rightarrow \text{Int}$ | Three: $(\text{Int}, \text{Int})$ , $(\text{Int}, \text{Float})$ , $(\text{Float}, \text{Float})$ |
| Overloaded?            | No                                                     | Arguably yes                                                                                      |

## CHAPTER 3

### LEXICAL ANALYSIS AND SYMBOL TABLE

This chapter covers the responsibilities related to lexical and early semantic foundations of the language. It includes defining token structures and categories (such as keywords, operators, identifiers, and literals), implementing the lexer to transform source characters into tokens, and establishing the symbol table to manage variable and function storage along with their associated types. Together, these components ensure that raw source input is systematically organised into meaningful structures that can be used reliably by later stages of the compiler.

#### 3.1 Token representation

The token system, written in `components/tokens.py`, establishes the vocabulary and structure of all lexical elements in the language. It specifies both the types of tokens that can exist and the structure each token must follow. This serves as a canonical contract between the lexer and the parser, ensuring consistency across all stages of the compiler.

Table 3.1 summarises the main categories, example lexemes, and typical `TokenType` values.

Table 3.1. Token categories and representative `TokenType` values

| Category          | Example lexemes                                                        | Representative <code>TokenType</code>                    |
|-------------------|------------------------------------------------------------------------|----------------------------------------------------------|
| Literals          | 42, 3.14, true, "hello"                                                | INT_LITERAL, FLOAT_LITERAL, BOOL_LITERAL, STRING_LITERAL |
| Identifiers       | x, counter, add                                                        | IDENTIFIER                                               |
| Keywords          | if, else, while, def, return                                           | keyword-specific token variants                          |
| Integer operators | +, -, *, /                                                             | PLUS, MINUS, STAR, SLASH                                 |
| Float operators   | +, -, *, /                                                             | PLUS_DOT, MINUS_DOT, STAR_DOT, SLASH_DOT                 |
| Other operators   | =, ==, !=                                                              | ASSIGN, EQUAL_EQUAL, BANG_EQUAL                          |
| Delimiters        | Parentheses, braces, comma, and semicolon (see lexer delimiter tokens) | delimiter token variants                                 |
| End marker        | end of source input                                                    | EOF                                                      |

Token types (from class `TokenType`) are defined using an Enum with `auto()` to automatically assign unique values. This avoids reliance on raw strings and ensures safer comparisons during parsing. The token categories include literals, identifiers, keywords, operators, delimiters,

and a special end-of-file (EOF) token. The EOF token is appended after all input is processed, allowing the parser to detect the end of input explicitly rather than encountering an unexpected termination.

Each token is represented as an immutable dataclass (`frozen=True`), meaning its fields cannot be modified after creation. Each token contains four key pieces of information: its kind (`token_type`), its lexeme (the exact substring from the source code), and its source position (`line`, `column`). This structured representation allows later stages to process the program in terms of syntax and errors without handling raw text directly. Immutability improves reliability by preventing accidental modification during later processing.

```
1 @dataclass(frozen=True)
2 class Token:
3     token_type: TokenType
4     lexeme: str
5     line: int
6     column: int
```

### 3.2 Lexer implementation

The lexer, implemented in `components/lexica.py`, is a hand-written scanner that converts raw source text into a sequence of tokens. Unlike generated lexers (for example SLY lexer mode, or Lex and Flex), this implementation uses explicit control flow (`if` and `while`) to process input.

The lexer maintains internal state including the source string, current position, start of the current token, line number, and column index. The column index is tracked per line and resets upon encountering a newline, while a separate `token_column` records the starting position of each token for accurate error reporting.

The main function, `tokenize`, iterates through the source code, repeatedly invoking `_scan_token` to identify the next token. Tokens are appended to a list, and an EOF token is added at the end.

Whitespace is skipped, while invalid input results in a `LexerError`.

```
1 def tokenize(self) -> list[Token]:
2     tokens: list[Token] = []
3     while not self._is_at_end():
4         self.start = self.current
5         self.token_column = self.column
6         token = self._scan_token()
7         if token is not None:
8             tokens.append(token)
9     tokens.append(Token(TokenType.EOF, "", self.line, self.column))
```

```
10     return tokens
```

Token recognition follows a structured approach. Single-character tokens are mapped using a lookup table. Multi-character operators such as `==` and `!=` are handled using lookahead via `_match`. String literals are processed by `_string`, which reads characters until a closing quotation mark or the end of input is reached, with error handling for unterminated strings. Numeric literals are handled by `_number`, which distinguishes integers and floating-point values based on the presence of a decimal point followed by digits. Identifiers are processed greedily, consuming alphanumeric characters and underscores, and then checked against the `KEYWORDS` map to determine whether they are reserved words or user-defined names. Notably, `true` and `false` are also entries in this map, mapping to `TokenType.BOOL_LITERAL` rather than a keyword-specific type. This means boolean literals are recognised through the same keyword-fallback mechanism as reserved words, preventing them from being scanned as user-defined identifiers. In contrast, `print` is *not* reserved: it is scanned as an ordinary `IDENTIFIER` and later treated as a built-in callable by the type checker and interpreter.

```
1 KEYWORDS: dict[str, TokenType] = {
2     "if": TokenType.IF,
3     "else": TokenType.ELSE,
4     "while": TokenType.WHILE,
5     # ...
6     "true": TokenType.BOOL_LITERAL,
7     "false": TokenType.BOOL_LITERAL,
8 }

1 # Arithmetic operators with optional trailing '.' for float variant
2 _ARITH_DOT: dict[str, tuple[TokenType, TokenType]] = {
3     "+": (TokenType.PLUS, TokenType.PLUS_DOT),
4     "-": (TokenType.MINUS, TokenType.MINUS_DOT),
5     "*": (TokenType.STAR, TokenType.STAR_DOT),
6     "/": (TokenType.SLASH, TokenType.SLASH_DOT),
7 }
```

When one of these characters is encountered, the lexer checks the next character: if it is a `.` (and the character after that is not a digit), the dot-suffixed float variant is emitted; otherwise the plain integer operator is emitted. This lookahead prevents the `.` in a float literal such as `3.14` from being consumed as part of an operator.

```
1 SINGLE_CHAR_TOKENS: dict[str, TokenType] = {
2     "=": TokenType.ASSIGN,
3     "(": TokenType.LPAREN,
4     ")": TokenType.RPAREN,
5     # ...
```

Helper functions such as `_advance`, `_peek`, and `_peek_next` manage character consumption and lookahead, while `_is_at_end` determines when input has been fully processed. Error messages include precise line and column information, aiding debugging and diagnostics.

Table 3.2 lists the main functions and data structures in `components/lexica.py` and their roles.

Table 3.2. Key functions and data in `lexica.py`

| Function / data                                                        | Responsibility                                          | Output / effect             |
|------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------|
| <code>tokenize()</code>                                                | Iterates through source, scans tokens, appends EOF      | ordered token list          |
| <code>_scan_token()</code>                                             | Dispatches recognition based on character and lookahead | next token or skip          |
| <code>_number()</code>                                                 | Parses numeric sequences, distinguishes int vs float    | numeric literal token       |
| <code>_string()</code>                                                 | Parses quoted text, checks termination                  | string token or error       |
| <code>_identifier()</code>                                             | Parses identifiers, resolves keyword vs user-defined    | keyword or identifier token |
| KEYWORDS map                                                           | Maps reserved words and Boolean literals to token types | token type selection        |
| SINGLE_CHAR_TOKENS map                                                 | Maps single-character symbols                           | token type selection        |
| <code>_advance()</code> , <code>_peek()</code> , <code>_match()</code> | Character consumption and lookahead                     | scanner state update        |

### 3.3 Symbol table and semantic structure

The symbol table, defined in `components/symbol_table.py`, operates at a higher level than the lexer. While the lexer processes raw text into tokens, the symbol table manages semantic information such as variable and function names, their types, and their scope.

The symbol table is implemented as a scoped structure, where each scope maintains its own dictionary of symbols and optionally references a parent scope. This enables proper handling of nested blocks and variable shadowing. Symbols are introduced through definition functions and retrieved through lookup operations, which search recursively through parent scopes if needed.

```

1 def lookup(self, name: str) -> Symbol:
2     symbol = self._symbols.get(name)
3     if symbol is not None:
4         return symbol
5     if self.parent is not None:

```

```

6         return self.parent.lookup(name)
7         raise SymbolTableError(f"Undefined symbol: {name}")

```

The type system is represented using a `LanguageType` enumeration, which includes fundamental types such as `Integer`, `Float`, `Boolean`, `String`, and `Void`. Symbols are modelled using an inheritance hierarchy: the base `Symbol` class stores a name and type, while subclasses such as `VariableSymbol` and `FunctionSymbol` include additional attributes. Variables track initialization state, while functions store parameter lists and return types. Table 3.3 summarises the main building blocks.

Table 3.3. Symbol table components and roles

| Component                      | Stored information                                                                                         | Role in semantic analysis                                 |
|--------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Scoped symbol table            | Symbols in current and parent scope                                                                        | Supports nesting and shadowing via recursive lookup       |
| <code>LanguageType</code> enum | <code>Integer</code> , <code>Float</code> , <code>Boolean</code> , <code>String</code> , <code>Void</code> | Defines the type system                                   |
| <code>Symbol</code> (base)     | Name and declared or inferred type                                                                         | Base abstraction for entries                              |
| <code>VariableSymbol</code>    | Variable metadata (including initialization state)                                                         | Tracks correct usage before assignment                    |
| <code>FunctionSymbol</code>    | Parameters and return type                                                                                 | Validates calls and return consistency                    |
| <code>lookup(name)</code>      | Recursive scope search                                                                                     | Resolves identifiers or reports undefined                 |
| <code>SymbolTableError</code>  | Semantic error details                                                                                     | Reports duplicates, undefined symbols, invalid operations |

Errors related to semantic violations are handled via `SymbolTableError`. These include duplicate declarations within the same scope, references to undefined symbols, and invalid operations such as attempting to mark a non-variable symbol as initialized.

### 3.4 Role of the symbol table in the pipeline

The symbol table is constructed during the type-checking phase rather than during lexical analysis. This guideline is followed in the implementation. The type checker initialises a global symbol table and populates it while traversing the AST, including defining variables and functions, managing nested scopes, and validating type correctness.

```

1 def run_pipeline(source_code: str) -> PipelineResult:
2     tokens = Lexer(source_code).tokenize()
3     tree = Parser(tokens).parse()
4     parse_tree_text = ParseTreePrinter().print(tree)
5     ast_text = ASTPrinter().print(tree)

```

```

6     checked_scope = TypeChecker().check(tree)
7     outputs: list[str] = []
8     Interpreter(output_callback=outputs.append).execute(tree)
9     return PipelineResult(
10         tokens=tokens,
11         parse_tree_text=parse_tree_text,
12         ast_text=ast_text,
13         checked_scope=checked_scope,
14         outputs=outputs,
15     )

```

Once type checking is complete, the populated symbol table is returned as part of the pipeline result (`checked_scope`). The checker creates child scopes for blocks and functions during analysis, but the displayed `checked_scope` table represents the global scope entries returned by the pipeline. It can then be displayed or used for further analysis.

This separation of concerns keeps lexical analysis focused on tokenization, while semantic meaning is enforced during type checking. For user-defined functions, definitions are first registered in the global scope with placeholder parameter and return types. These are inferred from the first valid call and refined through body checking; subsequent calls must match the inferred parameter types.

## CHAPTER 4

# RECURSIVE-DESCENT PARSING, PARSE TREE, AND ABSTRACT SYNTAX TREE

This chapter presents the implementation of the recursive-descent parser, the grammar-oriented parse-tree view exported for inspection, and the associated abstract syntax tree (AST) model used for downstream semantic analysis and execution. Starting from lexer-produced tokens, the parser constructs a hierarchical Program AST. A separate `ParseTreePrinter` then renders that parsed program in an ASCII branch-style rule-oriented tree form so that the submission can show explicit correspondence between grammar productions and accepted programs without changing the executable internal representation. Implemented syntax coverage includes arithmetic expressions, assignments, if-then-else, while, function definitions, function calls, return, and built-in `print(...)` calls. Equality operators (`==`, `!=`) are parsed as ordinary lowest-precedence expressions, so Boolean results can appear in conditions, assignments, function arguments, and `print(...)` expressions.

### 4.1 Abstract syntax tree representation

The AST is defined as a collection of Python dataclasses representing the structural elements of the programming language. It serves as an intermediate representation between parsing and later stages such as type checking and interpretation.

A key distinction exists between tokens and AST nodes. Tokens are flat and describe individual lexical elements such as operators or identifiers. In contrast, the AST is hierarchical. For example, while a token may represent the symbol `+`, a `BinaryOp` node represents an entire addition operation, including its left and right operands as subtrees.

All AST nodes inherit from a common base class, allowing them to store positional metadata (line, column). This ensures that errors detected in later stages can still reference precise locations in the original source code.

```
1 @dataclass
2 class ASTNode:
3     """Base class for every AST node. Carries source location for
4         diagnostics."""
5     line: int = field(default=0, kw_only=True)
```

```

5     column: int = field(default=0, kw_only=True)

```

### 4.1.1 Expressions

Expressions represent computations or values within the program. A `Literal` node stores values known at parse time, including integers, floating-point numbers, booleans, and strings. String literals are stored without surrounding quotation marks, as these are removed during parsing.

```

1 if tok.token_type == TokenType.STRING_LITERAL:
2     self._advance()
3     return Literal(value=tok.lexeme[1:-1], line=tok.line, column=tok.
        column)

```

An `Identifier` node represents the use of a variable name within an expression, indicating evaluation rather than declaration. A `BinaryOp` node models arithmetic and comparison operations such as addition, subtraction, multiplication, division, and equality checks, storing the operator along with references to left and right operands. A `FunctionCall` node represents function invocation, consisting of the function name and a list of argument expressions.

```

1 @dataclass
2 class BinaryOp(ASTNode):
3     """An arithmetic or comparison binary operation.
4
5     op is one of: '+' '-' '*' '/' '+.' '-.' '*.' '/.' '==' '!=',
6     Example:  a + b    x == 0    a +. b
7     """
8     op: str
9     left: ASTNode
10    right: ASTNode

```

### 4.1.2 Statements

Statements represent actions or control flow within the program. An `Assign` node models variable assignment, pairing a variable name with a value expression, while `Return` stores a single expression and represents function return. A `FunctionCall` node may also appear as a standalone statement, which is how built-in `print(...)` calls are represented after parsing. A `Block` node groups multiple statements within braces, representing a scoped execution sequence.

Control flow is handled via `If` and `While` nodes. The `If` node stores a condition, a then-block, and an optional else-block, while `While` stores a loop condition and body block. Function

definitions are represented by `FunctionDef` nodes, which include the function name, a list of parameter names, and a body block. Parameter types are intentionally not assigned at parse time, as type inference is handled later by the type checker. Finally, `Program` serves as the AST root, containing all top-level statements.

## 4.2 Parser implementation

The parser consumes the list of tokens generated by the lexer and produces a single `Program` AST node. It is implemented as a hand-written recursive-descent parser based on the project grammar. This avoids parser generators and instead relies on explicit functions for each grammar rule.

The parser maintains internal state consisting of the token list and a position index (`self._pos`) indicating the current token. Tokens are accessed via helper methods such as `_peek` and `_advance`. Parsing errors are reported using a custom `ParseError` exception with line and column information consistent with lexer diagnostics.

```
1 def parse(self) -> Program:
2     """Parse the full token stream and return the root Program node."""
3     line, col = self._peek().line, self._peek().column
4     statements: list[ASTNode] = []
5     while not self._is_at_end():
6         statements.append(self._statement())
7     return Program(statements=statements, line=line, column=col)
```

In addition to the main `parse()` entry point, the parser's statement dispatcher (`_statement`) implements a deterministic decision order over token patterns: function definition, conditional, loop, return, assignment, then expression statement fallback. This design is important because it prevents ambiguity between syntactically similar forms. In particular, assignments are recognized using one-token lookahead (`IDENTIFIER` followed by `=`), while identifier-led expressions that are not assignments are parsed as expression statements and must still end with a semicolon. This gives the parser full coverage of both declarative-style and expression-driven top-level statements without requiring a parser generator.

## 4.3 Parse-tree view

Although the parser's internal product is an AST, the project now also exposes a separate parse-tree view for reporting and demonstration. This view is generated by `components/parse_tree_printer.py` from the parsed program and uses grammar-oriented layers such as `program`, `statement`,

assignment, expr, additive, term, factor, function\_call, and args. Terminal-like leaves are rendered as IDENTIFIER(x), INT\_LITERAL(5), '(', ')', and EOF, with ASCII branch markers indicating parent-child structure.

This design keeps the semantic implementation simple: the type checker and interpreter still consume the AST, while the parse-tree renderer exists purely to expose the grammar structure required for inspection.

Table 4.1. Parse tree versus AST in the implemented system

| Artifact   | Purpose                          | Used by                   |
|------------|----------------------------------|---------------------------|
| Parse tree | Grammar-oriented inspection view | CLI, UI, report           |
| AST        | Executable semantic tree         | Type checker, interpreter |

For presentation purposes, Figure 4.1 gives a tiny textbook-style diagram for the source fragment `a = 3;.` This figure is included only as an explanatory visual in the report; the actual system output uses the more scalable ASCII branch representation shown in the CLI and UI.

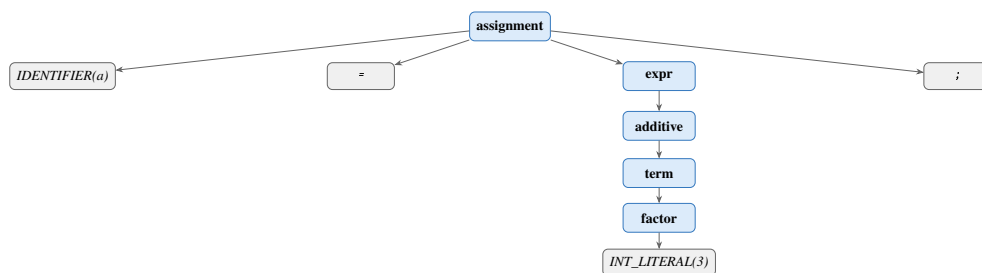


Figure 4.1. Tiny textbook-style parse-tree diagram for `a = 3;.`

### 4.3.1 Statement parsing

The main parsing loop repeatedly processes statements until EOF, with the statement type determined via dispatch on the current token. Function definitions, conditionals, loops, and returns each have dedicated parsing methods. Assignments are detected using one-token lookahead (IDENTIFIER followed by ASSIGN), preventing misclassification of function calls as assignments. If no specific statement pattern matches, the parser falls back to an expression statement and enforces a terminating semicolon.

```

1 # assignment: IDENTIFIER "=" expr ";"
2 if (
3     tok.token_type == TokenType.IDENTIFIER
4     and self._peek_next().token_type == TokenType.ASSIGN
5 ):
6     return self._assignment()
  
```

```

7 # fallback: bare expression statement  expr ";"
8 node = self._expr()
9 self._expect(TokenType.SEMICOLON, "Expected ';' after expression")

```

### 4.3.2 Expression parsing and precedence

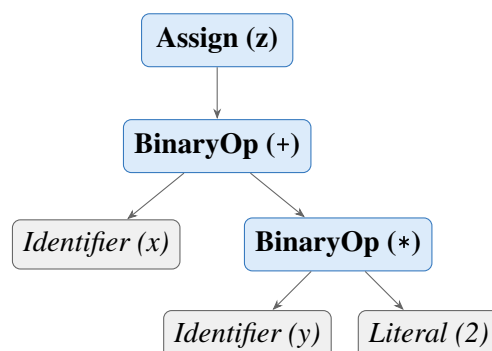
Expressions are parsed in layered precedence levels: `_expr` handles comparison operators first (`==`, `!=`) after parsing an additive left-hand side, then `_additive` handles `+`, `-`, `+. .`, and `-.`; `_term` handles `*`, `/`, `*.`, and `/.`; and `_factor` handles literals, identifiers and function calls, and parenthesized expressions. Function calls are recognised when an identifier is followed by `(`, with argument lists parsed as comma-separated expressions. Parenthesized expressions are supported to override precedence. Unary negation (`-x`, `-5`) is also handled at the `_factor` level: a leading `-` token causes the parser to recursively parse the operand and return `BinaryOp(0 - operand)`, so the existing type rules for subtraction cover negation without any additional inference code. Float unary negation is handled similarly for `-.`, producing `BinaryOp(0.0 - . operand)`.

```

1 def _expr(self) -> ASTNode:      # ==, !=
2     ...
3 def _additive(self) -> ASTNode:  # +, -, +. ., -.
4     ...
5 def _term(self) -> ASTNode:      # *, /, *. ., /.
6     ...
7 def _factor(self) -> ASTNode:    # unary -, literals, identifiers/calls,
8     ...                          grouped expr

```

Figure 4.2 shows the AST produced for the statement `z = x + y * 2;`. Because `_term` handles `*` before `_additive` handles `+`, the multiplication sub-tree is nested one level deeper, encoding the higher precedence of `*` over `+` without any explicit priority table at runtime.

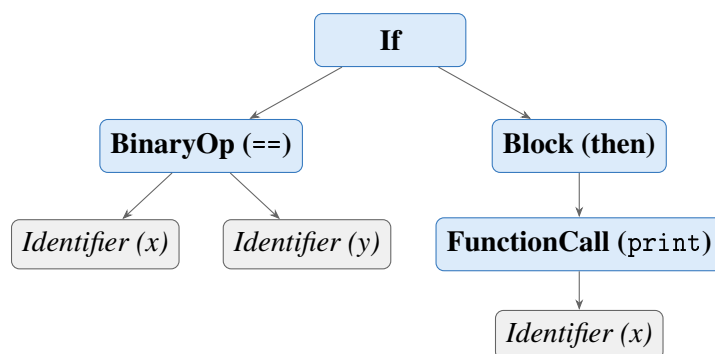


**Figure 4.2.** AST for `z = x + y * 2;`. Multiplication is a deeper sub-tree than addition, encoding higher precedence through tree structure rather than a runtime priority table.

This structural encoding of precedence eliminates the need for runtime operator-priority resolution and keeps semantic analysis independent of parsing strategy.

Expression parsing is intentionally split into layered methods to enforce precedence and associativity in a transparent way. Parenthesized expressions are parsed explicitly and re-enter `_expr`, allowing precedence override where required by source syntax. Comparison operators (`==`, `!=`) are parsed at the top of the `_expr` method as the lowest-precedence binary operators. Because comparisons are part of the general expression grammar, Boolean results can appear in any expression context — as conditions, assignments, arguments, or print expressions. Semantic enforcement that a condition is truly Boolean is deferred to the type checker.

Figure 4.3 shows the AST for an `if` statement with a comparison condition, illustrating how the parser structures control flow with a Boolean sub-tree.



**Figure 4.3.** AST for `if (x == y) { print(x); }`. The comparison produces a `BinaryOp` node whose inferred type is Boolean, checked by the type checker before execution.

### 4.3.3 Helper functions and lookahead

As described above, the parser uses `_check`, `_expect`, `_peek`, `_peek_next`, `_advance`, and `_is_at_end` for token navigation and validation. Lookahead is central for ambiguity resolution: assignment versus function call is resolved by inspecting the next token, while comparison operators are recognised in `_expr` after parsing the left additive expression. Parser errors follow a consistent diagnostic format: `[line X, col Y] ParseError: <message>`.

The parser also includes dedicated list parsers for formal parameters (`_params`) and call arguments (`_args`), both implemented as comma-separated sequences with optional emptiness. This ensures consistent handling of signatures such as `def f()` and `def f(a, b)` as well as invocations like `f()` and `f(x, y+1)`.

## 4.4 AST printing

For debugging and reporting, an AST printer converts the tree into a readable indented string, integrated into the pipeline output. The printer uses visitor-style dynamic dispatch, mapping node class names to methods such as `_visit_If` and `_visit_BinaryOp`, with output built as multi-line text where indentation reflects tree depth. Node formatting emphasises structure: `Program` and `Block` include statement counts, `BinaryOp` explicitly shows left and right subtrees, and `FunctionCall` prints each argument by index. `Literal` includes both value and runtime Python type names for clarity.

## 4.5 Parser and type checker boundary

A deliberate architectural boundary is maintained between syntax construction and semantic validation. The parser is responsible for producing a structurally valid AST with accurate source locations. The parse-tree renderer exposes the same parsed program in grammar-facing form, but it is not used for execution. Type correctness and semantic constraints are checked later by the type checker. For example, `_expr` may syntactically produce a node whose eventual type is not `Boolean`, but rejection of non-`Boolean` control-flow conditions is handled during semantic analysis. This separation keeps parser logic focused on grammar conformance and allows a clear responsibility split across team member contributions.

## CHAPTER 5

### TYPE INFERENCE ALGORITHM

#### 5.1 Overview

Type inference in this project means that the programmer does not write explicit type declarations for variables or functions. Instead, the compiler infers types from assignments, literals, operations, function calls, and return statements. This keeps the language simple while still preserving static type safety.

The type checker runs after parsing. It receives the AST and validates the program before execution. Any inconsistent use of types is reported as a type error, and execution is stopped.

The interpreter stage does not repeat this validation: it walks the same AST with ordinary Python representations of values. Static type information remains in the symbol table produced by the checker; run-time values are not separately labelled with `LanguageType`, except that `print()` formats output by inspecting the value's Python type.

#### 5.2 Type Environment

The type environment is represented by the symbol table. Conceptually, it is a mapping

$$\Gamma : \text{Identifier} \rightarrow \text{Type}$$

The environment grows as the checker visits the AST. On first assignment, a variable name is inserted with its inferred type. When a new block or a function call is entered, a child environment is created so that nested scopes can be checked independently while still accessing outer bindings when needed.

#### 5.3 Type Inference Procedure

The type checker performs type inference by traversing the abstract syntax tree before interpretation. The algorithm is syntax-directed: each AST node is visited, the types of its child nodes are inferred first, and the current node is checked according to the language typing rules.

**Algorithm** TYPECHECK(*node*,  $\Gamma$ )

**Input:** *node* = current AST node;  $\Gamma$  = current type environment / symbol table

**Output:** inferred type of *node*, or `TypeError`

1. **If** *node* is a **Literal**: return the corresponding primitive type: `Integer`, `Float`, `Boolean`, or `String`.
2. **If** *node* is an **Identifier**: look up the identifier in  $\Gamma$ . If it is not found, report an undefined variable error. Otherwise, return its stored type.
3. **If** *node* is a **BinaryOp**:
  - Infer the type of the left operand and the right operand.
  - If the operator is `+`, `-`, `*`, or `/`: require both operands to be `Integer`; return `Integer`.
  - If the operator is `+`, `-`, `*`, or `/`: require both operands to be `Float`; return `Float`.
  - If the operator is `==` or `!=`: require both operands to be the same arithmetic type; return `Boolean`.
  - Otherwise, report a `TypeError`.
4. **If** *node* is an **Assignment**:
  - Infer the type of the right-hand-side expression.
  - If the variable is not yet in  $\Gamma$ : insert it with the inferred type.
  - Else: verify the new type matches the existing type; if not, report a `TypeError`.
5. **If** *node* is an **If** statement:
  - Infer the condition type; require it to be `Boolean`.
  - Type-check the then-block in a child environment.
  - If an else-block exists, type-check it in a child environment.
6. **If** *node* is a **While** statement:
  - Infer the condition type; require it to be `Boolean`.
  - Type-check the loop body in a child environment.
7. **If** *node* is a **FunctionDef**:
  - Register the function name in  $\Gamma$ .
  - Parameter types are initially unknown.
  - Return type is inferred from return statements in the body.

8. If *node* is a **FunctionCall**:

- Infer all argument types.
- If this is the first valid call: store the argument types as the function’s parameter types.
- Else: compare argument types with the stored parameter types.
- Type-check the function body using a local environment.
- Return the inferred function return type.

9. If *node* is a **FunctionCall**:

- If the function is the built-in `print`: require exactly one argument, infer that argument type, and return `Void`.
- Otherwise, infer all argument types, validate them against the function signature, type-check the callee body if needed, and return the inferred function return type.

## 5.4 Inference Rules

### 5.4.1 Literals

Literal nodes have direct types:

$$\Gamma \vdash 42 : \text{Integer}$$
$$\Gamma \vdash 3.14 : \text{Float}$$
$$\Gamma \vdash \text{true} : \text{Boolean}$$
$$\Gamma \vdash \text{“hi”} : \text{String}$$

### 5.4.2 No Operator Overloading

The language does not support operator overloading. Following the Caml Light convention, integer and float arithmetic use entirely separate operator tokens, so every operator has a single, fixed type signature:

- `+`, `-`, `*`, `/` — require two `Integer` operands; result is `Integer`.
- `+. , -. , *. , /. ,` — require two `Float` operands; result is `Float`.
- `+` is not overloaded for string concatenation; applying it to a `String` is a type error.

- `==` and `!=` are not overloaded for Boolean or String equality.
- No implicit numeric promotion: `Integer + Float` is a type error.

Because every operator has a unique, unambiguous type derivable from its token alone, the compiler can always infer expression types without requiring explicit type declarations from the programmer. This design ensures that operator meaning is determined uniquely from the operator token itself, eliminating semantic ambiguity during type inference and simplifying static analysis.

### 5.4.3 Arithmetic Expressions

The integer operators `+`, `-`, `*`, and `/` require both operands to be `Integer` and always return `Integer` (including division, which uses integer floor division). The float operators `+. , - . , * . ,` and `/ .` require both operands to be `Float` and always return `Float`. Mixing an `Integer` and a `Float` with any operator is a type error.

Table 5.1. Arithmetic operator type rules

| Left operand | Operator                                                          | Right operand     | Result type |
|--------------|-------------------------------------------------------------------|-------------------|-------------|
| Integer      | <code>+</code> , <code>-</code> , <code>*</code> , <code>/</code> | Integer           | Integer     |
| Float        | <code>+. , - . , * . , / .</code>                                 | Float             | Float       |
| Integer      | any                                                               | Float             | TypeError   |
| Float        | any                                                               | Integer           | TypeError   |
| Any          | any                                                               | String or Boolean | TypeError   |

The following formal typing judgements capture these rules precisely. For integer operators ( $\text{op} \in \{+, -, *, /\}$ ):

$$\frac{\Gamma \vdash e_1 : \text{Integer} \quad \Gamma \vdash e_2 : \text{Integer}}{\Gamma \vdash e_1 \text{ op } e_2 : \text{Integer}}$$

For float operators ( $\text{op} \in \{+. , - . , * . , / .\}$ ):

$$\frac{\Gamma \vdash e_1 : \text{Float} \quad \Gamma \vdash e_2 : \text{Float}}{\Gamma \vdash e_1 \text{ op } e_2 : \text{Float}}$$

For mixed-type operands with any operator (and symmetrically for *Float/Integer*):

$$\frac{\Gamma \vdash e_1 : \text{Integer} \quad \Gamma \vdash e_2 : \text{Float}}{\text{TypeError}}$$

Note that each rule is keyed on the operator token, not on the operand types. This is only

possible because integer and float arithmetic use distinct tokens — the property that eliminates operator overloading.

### ***Unary negation***

Unary integer negation ( $-x$ ) is desugared by the parser to  $0 - x$ . Unary float negation ( $-.x$ ) is desugared to  $0.0 - .x$ . The type checker therefore applies the standard binary rules in both cases.

### ***5.4.4 Boolean Expressions***

The comparison operators `==` and `!=` always produce `Boolean`. Both operators require two operands of the *same* arithmetic type — either both `Integer` or both `Float`; mixed-type comparisons such as `Integer == Float` are rejected. Comparisons involving `Boolean` or `String` operands are also rejected by the type checker. Although `Boolean` and `String` values exist in the language, equality and inequality are deliberately restricted to arithmetic expressions because the project specification defines basic `Boolean` expressions as equality and inequality between two arithmetic expressions.

Furthermore, the same-type requirement for comparisons — both operands must be either both `Integer` or both `Float` — is a direct consequence of the no-overloading design. If `==` accepted mixed `Integer` and `Float` operands, the compiler would need to determine whether to compare by promotion (implicitly converting `Integer` to `Float`) or by value identity, introducing the same ambiguity that the separate-operator design eliminates for arithmetic. Rejecting mixed-type comparisons ensures that `==` and `!=` each have exactly two valid type signatures —  $(Integer, Integer) \rightarrow Boolean$  and  $(Float, Float) \rightarrow Boolean$  — rather than a third mixed-type signature.

### ***5.4.5 Assignment Statement***

Assignments define variable types on first use. If a variable does not yet exist in the type environment, the checker infers the type of the right-hand side and inserts the variable into the environment. If the variable already exists, the new value must have the same type.

If  $x \notin \Gamma$  and  $\Gamma \vdash e : T$ , then  $\Gamma, x : T \vdash x = e$

If  $x : T \in \Gamma$  and  $\Gamma \vdash e : T$ , then  $\Gamma \vdash x = e$

### 5.4.6 If-Then-Else

The condition of an if statement must be Boolean. The then-block and else-block are checked in child scopes so that local operations can be validated cleanly.

$$\Gamma \vdash c : \text{Boolean} \quad \Gamma' \vdash B_{\text{then}} \quad \Gamma'' \vdash B_{\text{else}} \implies \Gamma \vdash \text{if}(c) B_{\text{then}} \text{ else } B_{\text{else}}$$

### 5.4.7 While-Loop

The condition of a while loop must also be Boolean. The loop body is checked in a child scope and must not violate the types of variables already introduced in the surrounding environment.

$$\Gamma \vdash c : \text{Boolean} \quad \Gamma' \vdash B \implies \Gamma \vdash \text{while}(c) B$$

### 5.4.8 Function Definition and Call

Function parameter types are inferred from the first call to the function. After that first call, subsequent calls must use arguments of the same types. The return type is inferred from the return statements in the function body. If the function returns inconsistent types, the checker reports an error.

For functions that are defined but never called, the type checker performs a post-pass after all top-level statements have been processed. It infers parameter types by scanning the body for arithmetic and comparison usage, then checks the full body with those inferred types. This ensures that type errors inside never-called function bodies are still detected before execution.

Recursive calls are intentionally outside the scope of this implementation; supporting them would require additional fixed-point inference or explicit recursion handling that is orthogonal to the core type-inference algorithm.

## 5.5 Type Error Reporting

Type errors are reported with source positions so that the user can locate the problem quickly. The format used by the system is:

```
[line X, col Y] TypeError: message
```

Examples include assigning a String to an Integer variable, using a non-Boolean condition in an if statement, or supplying the wrong argument type to a function.

## 5.6 Example Type Inference Traces

Consider the following program fragment:

```
1 x = 3;  
2 y = 4;  
3 z = x + y;  
4 print(z);
```

The checker infers:

- `x`: Integer (from literal 3)
- `y`: Integer (from literal 4)
- `x + y`: Integer (Integer + Integer  $\rightarrow$  Integer)
- `z`: Integer

For a float example using dot-suffixed operators:

```
1 def addf(a, b) {  
2     return a +. b;  
3 }  
4  
5 result = addf(2.0, 3.5);
```

The first function call infers `a`: Float and `b`: Float. The operator `+.`  requires both operands to be Float, so the function return type is inferred as Float. The variable `result` is also inferred as Float.

## CHAPTER 6

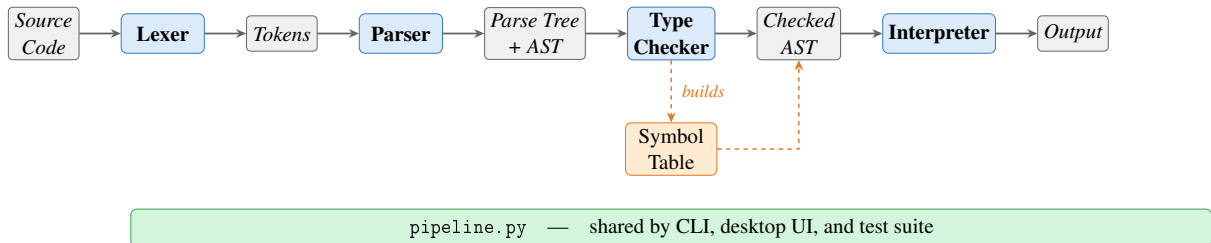
### SYSTEM ARCHITECTURE AND IMPLEMENTATION

This chapter describes how the implementation is organised: the end-to-end processing flow, repository layout, technology choices, and the main components that realise static analysis and execution (type checker, interpreter, and user interfaces). Detailed treatment of lexical analysis, the symbol table, and parsing appears in Chapters 3 and 4; Chapter 5 presents the type-inference rules that the type checker implements.

#### 6.1 Compiler pipeline

The system follows a pipeline with four processing stages and five labelled output sections exposed by the shared pipeline. Lexical and syntactic processing are kept separate from semantic analysis and execution, which simplifies testing and localises errors to the appropriate stage.

The overall flow of data through the system is shown in Figure 6.1.



**Figure 6.1. Front-end and interpreter pipeline overview.**

Source characters are scanned by the lexer into a token stream. The parser consumes tokens and builds an AST that reflects program structure (expressions, statements, and control flow). The shared pipeline also renders a separate parse-tree view from the parsed program so that grammar structure can be inspected explicitly in the CLI, UI, and report. The type checker then walks the AST, populates and uses the symbol table, infers types, and validates scope and type rules. Only programs that pass the type checker proceed to interpretation. The interpreter evaluates the checked tree using ordinary Python values; it does not re-run the type rules, since correctness at runtime is guaranteed by the prior static pass. All stages are orchestrated through `pipeline.py`, which is shared by the command-line runner, the desktop UI, and the test suite.

### 6.1.1 End-to-end pipeline walkthrough

The following example traces the complete execution of a five-statement program through the exported pipeline views and execution stages, illustrating the correspondence between source text, tokens, parse tree, abstract syntax tree, inferred types, and runtime output.

#### Source program:

```
1 a = 3;
2 b = 4;
3 print(a + b);
4 c = 1.5;
5 print(c +. c);
```

#### Stage 1 — Lexer output (token stream):

|    |               |         |                |
|----|---------------|---------|----------------|
| 1  | IDENTIFIER    | 'a'     | line 1, col 1  |
| 2  | ASSIGN        | '='     | line 1, col 3  |
| 3  | INT_LITERAL   | '3'     | line 1, col 5  |
| 4  | SEMICOLON     | ';'     | line 1, col 6  |
| 5  | IDENTIFIER    | 'b'     | line 2, col 1  |
| 6  | ASSIGN        | '='     | line 2, col 3  |
| 7  | INT_LITERAL   | '4'     | line 2, col 5  |
| 8  | SEMICOLON     | ';'     | line 2, col 6  |
| 9  | IDENTIFIER    | 'print' | line 3, col 1  |
| 10 | LPAREN        | '('     | line 3, col 6  |
| 11 | IDENTIFIER    | 'a'     | line 3, col 7  |
| 12 | PLUS          | '+'     | line 3, col 9  |
| 13 | IDENTIFIER    | 'b'     | line 3, col 11 |
| 14 | RPAREN        | )'      | line 3, col 12 |
| 15 | SEMICOLON     | ';'     | line 3, col 13 |
| 16 | IDENTIFIER    | 'c'     | line 4, col 1  |
| 17 | ASSIGN        | '='     | line 4, col 3  |
| 18 | FLOAT_LITERAL | '1.5'   | line 4, col 5  |
| 19 | SEMICOLON     | ';'     | line 4, col 8  |
| 20 | IDENTIFIER    | 'print' | line 5, col 1  |
| 21 | LPAREN        | '('     | line 5, col 6  |
| 22 | IDENTIFIER    | 'c'     | line 5, col 7  |
| 23 | PLUS_DOT      | '+. '   | line 5, col 9  |
| 24 | IDENTIFIER    | 'c'     | line 5, col 12 |
| 25 | RPAREN        | )'      | line 5, col 13 |
| 26 | SEMICOLON     | ';'     | line 5, col 14 |
| 27 | EOF           |         | line 5, col 15 |

#### Stage 2 — Parser output (ASCII branch parse tree):

```
1 program
2 |-- statement
3 |   '-- assignment
4 |       |-- IDENTIFIER(a)
5 |       |-- '='
6 |       |-- expr
```

```

7 |         |         '-- additive
8 |         |         '-- term
9 |         |         '-- factor
10 |        |         '-- INT_LITERAL(3)
11 |        '-- ';'
12 |-- statement
13 |   '-- assignment
14 |       |-- IDENTIFIER(b)
15 |       |-- '='
16 |       |-- expr
17 |           '-- additive
18 |           '-- term
19 |           '-- factor
20 |           '-- INT_LITERAL(4)
21 |       '-- ';'
22 |-- statement
23 |   '-- expr_stmt
24 |       |-- expr
25 |           '-- additive
26 |           '-- term
27 |           '-- factor
28 |           '-- function_call
29 |               |-- IDENTIFIER(print)
30 |               |-- '('
31 |               |-- args
32 |                   '-- expr
33 |                       '-- additive
34 |                           |-- additive
35 |                               '-- term
36 |                                   '-- factor
37 |                                       '-- IDENTIFIER(a)
38 |                                           |-- '+'
39 |                                               '-- term
40 |                                                   '-- factor
41 |                                                       '-- IDENTIFIER(b)
42 |               '-- ')
43 |       '-- ';'
44 |-- statement
45 |   '-- assignment
46 |       |-- IDENTIFIER(c)
47 |       |-- '='
48 |       |-- expr
49 |           '-- additive
50 |           '-- term
51 |           '-- factor
52 |           '-- FLOAT_LITERAL(1.5)
53 |       '-- ';'
54 |-- statement
55 |   '-- expr_stmt
56 |       |-- expr
57 |           '-- additive
58 |           '-- term
59 |           '-- factor

```

```
60 |         '-- function_call  
61 |             |-- IDENTIFIER(print)  
62 |             |-- '(',  
63 |             |-- args  
64 |                 '-- expr  
65 |                     '-- additive  
66 |                         |-- additive  
67 |                             |-- term  
68 |                                 |-- factor  
69 |                                     '-- IDENTIFIER(c)  
70 |                                         |-- '+.'  
71 |                                             '-- term  
72 |                                                 '-- factor  
73 |                                                     '-- IDENTIFIER(c)  
74 |                                                         '-- ')',  
75 |         '-- ';' ,  
76 |     '-- EOF
```

### Stage 3 — Parser output (abstract syntax tree):

```

1 Program [5 statement(s)]
2     Assign 'a' =
3         Literal 3 (int)
4     Assign 'b' =
5         Literal 4 (int)
6     FunctionCall 'print' [1 arg(s)]
7         arg[0]:
8             BinaryOp '+'
9             left:
10                 Identifier 'a'
11             right:
12                 Identifier 'b'
13     Assign 'c' =
14         Literal 1.5 (float)
15     FunctionCall 'print' [1 arg(s)]
16         arg[0]:
17             BinaryOp '+.'
18             left:
19                 Identifier 'c'
20             right:
21                 Identifier 'c'

```

### Stage 4 — Type checker output (inferred type environment):

| Name | Inferred type |
|------|---------------|
| a    | Integer       |
| b    | Integer       |
| c    | Float         |

The symbol table records only named symbols (a, b, and c). The type checker additionally

infers that the sub-expression `a + b` has type `Integer` (`Integer + Integer → Integer`) and that `c +. c` has type `Float` (`Float +. Float → Float`) during AST traversal, but sub-expression types are not stored in the symbol table.

### Stage 5 — Interpreter output:

```
1 7 : Integer
2 3.0 : Float
```

The example demonstrates the full data flow: source characters are tokenised by the lexer, structured by the parser into both an inspectable parse tree and an executable AST, validated and annotated by the type checker, and finally evaluated by the interpreter to produce typed output. It also shows that the ASCII branch parse tree preserves the concrete operator spellings `'+'` and `'+. '`, making the distinction between integer and float arithmetic visible in the exported parser view.

### *Float operator walkthrough*

To illustrate how the dot-suffixed float operators flow through each pipeline stage, consider a second example:

### Source program:

```
1 x = 2.0;
2 y = 3.5;
3 print(x +. y);
```

### Stage 1 — Lexer output (token stream):

|    |               |         |               |
|----|---------------|---------|---------------|
| 1  | IDENTIFIER    | 'x'     | line=1 col=1  |
| 2  | ASSIGN        | '='     | line=1 col=3  |
| 3  | FLOAT_LITERAL | '2.0'   | line=1 col=5  |
| 4  | SEMICOLON     | ';'     | line=1 col=8  |
| 5  | IDENTIFIER    | 'y'     | line=2 col=1  |
| 6  | ASSIGN        | '='     | line=2 col=3  |
| 7  | FLOAT_LITERAL | '3.5'   | line=2 col=5  |
| 8  | SEMICOLON     | ';'     | line=2 col=8  |
| 9  | IDENTIFIER    | 'print' | line=3 col=1  |
| 10 | LPAREN        | '('     | line=3 col=6  |
| 11 | IDENTIFIER    | 'x'     | line=3 col=7  |
| 12 | PLUS_DOT      | '+. '   | line=3 col=9  |
| 13 | IDENTIFIER    | 'y'     | line=3 col=12 |
| 14 | RPAREN        | ')'     | line=3 col=13 |
| 15 | SEMICOLON     | ';'     | line=3 col=14 |
| 16 | EOF           |         | line=3 col=15 |

Note that the lexer emits `PLUS_DOT` (not `PLUS`) for `+.` , distinguishing float addition from integer

addition at the token level. Note also that `print` is tokenised as an ordinary `IDENTIFIER` — it is not a reserved keyword in this language.

### Stage 3 — Parser output (abstract syntax tree):

```
1 Program [3 statement(s)]
2   Assign 'x' =
3     Literal 2.0 (float)
4   Assign 'y' =
5     Literal 3.5 (float)
6   FunctionCall 'print' [1 arg(s)]
7     arg[0]:
8       BinaryOp '+.'
9         left:
10           Identifier 'x'
11         right:
12           Identifier 'y'
```

The AST records the operator as `+.`  rather than `+`, preserving the type distinction through to semantic analysis.

### Stage 4 — Type checker output (inferred type environment):

| Name | Inferred type |
|------|---------------|
| x    | Float         |
| y    | Float         |

The type checker verifies that both operands of `+.`  are `Float`. Had either operand been `Integer`, a `TypeError` would be reported before execution.

### Stage 5 — Interpreter output:

```
1 5.5 : Float
```

This second walkthrough demonstrates that the dot-suffixed operators are visible at every pipeline stage — from the lexer token `PLUS_DOT` to the AST node `BinaryOp '+. '` to the final typed output — ensuring that integer and float arithmetic are kept strictly separate throughout the system.

#### 6.1.2 Entry points and shared pipeline

All stages are orchestrated by `components/pipeline.py`, which is shared by the command-line runner, the desktop UI, and the test suite. From the project root, the CLI and desktop UI can be launched as follows:

```

1 # macOS / Linux
2 python3 main.py          # run CLI (built-in sample or script file)
3 python3 ui.py            # open desktop workbench
4
5 # Windows
6 .\venv\Scripts\python.exe main.py
7 .\venv\Scripts\python.exe ui.py

```

The CLI prints five labelled stages in order: tokens, parse tree, AST text, type table, and execution output. The desktop workbench uses the same pipeline internally, presenting each stage in a separate tab for inspection.

## 6.2 Project structure

The main source files and their responsibilities are summarised in Table 6.1.

Table 6.1. Main source files

| File                             | Responsibility                                       |
|----------------------------------|------------------------------------------------------|
| main.py                          | Command-line runner for sample code or a script file |
| ui.py                            | Desktop UI for editing and running a script          |
| components/tokens.py             | Token classes and token categories                   |
| components/lexica.py             | Lexical analysis                                     |
| components/ast_nodes.py          | AST node definitions                                 |
| components/parser.py             | Recursive-descent parser                             |
| components/parse_tree_printer.py | Parse-tree renderer for inspection                   |
| components/ast_printer.py        | AST renderer for debugging and reporting             |
| components/symbol_table.py       | Scoped symbol table and semantic types               |
| components/type_checker.py       | Static type checker and inference                    |
| components/interpreter.py        | Tree-walking interpreter                             |
| components/pipeline.py           | Shared pipeline for CLI, UI, and tests               |

## 6.3 Technology stack

The project uses Python and standard-library tools only (Table 6.2).

Table 6.2. Technology stack

| Tool         | Purpose                           |
|--------------|-----------------------------------|
| Python 3.10+ | Core implementation language      |
| dataclasses  | AST and runtime helper structures |
| tkinter      | Desktop graphical interface       |
| unittest     | Automated testing                 |

## 6.4 Type checker

The `TypeChecker` class implements the inference and checking rules described in Chapter 5. It traverses the AST before execution, uses the symbol table to record variable and function information, infers types on first assignment, validates arithmetic and comparison expressions, enforces Boolean conditions for control flow, infers function parameter types from the first call (or from body usage for uncalled functions), and infers function return types from `return` statements. Programs that fail these rules are rejected before interpretation.

### 6.4.1 Scope of the implemented type system

In scope terms, this component turns a syntactically valid AST into a semantically valid program:

- static typing rules for arithmetic, comparison, assignment, control flow, and functions,
- type checking for expressions and statements,
- variable type inference from first assignment,
- function parameter inference from the first call (or from body usage for uncalled functions),
- function return type inference from `return` statements,
- source-positioned type errors when a program violates a typing rule.

## 6.5 Interpreter

The interpreter is a tree-walking evaluator over the checked AST. It maintains nested runtime environments for variables and function calls. Assignments write through the current or nearest enclosing scope; `if` selects a branch; `while` repeats until its condition becomes false; function calls bind arguments by value and return through an internal return mechanism.

The built-in `print()` function outputs both value and runtime type. All four types are supported:

```
1 print(42);           (* 42 : Integer      *)
2 print(3.14);         (* 3.14 : Float       *)
3 print(true);         (* true : Boolean    *)
4 print("plc");        (* "plc" : String   *)
```

### 6.5.1 Scope of the implemented runtime

The runtime stage provides observable behaviour and integrates with the shared pipeline:

- assignment execution,
- if execution,
- while execution,
- function execution with value parameter passing,
- `print()` execution with immediate output,
- execution behind the same `run_pipeline` path used by the CLI, UI, and tests.

## 6.6 Graphical user interface

The desktop workbench (`ui.py`) is implemented with Python's `tkinter` standard library and requires no additional dependencies. The window is titled *125970-126690-126687-Ass2-PLC* and opens at  $1200 \times 760$  pixels in a horizontal split layout.

**Toolbar.** A top toolbar provides three buttons — *Open Script*, *Run*, and *Clear Output* — and a right-aligned status label that reports *Ready*, a loaded filename, or *Run failed*.

**Left panel (source editor).** A monospaced `Text` widget with both horizontal and vertical scrollbars allows the user to enter or paste arbitrary source code. The *Open Script* button populates the editor from a file. The editor is pre-loaded with the built-in sample program on first launch.

**Right panel (output tabs).** A `Notebook` widget presents six tabs, each backed by a scrollable read-only `Text` widget with both horizontal and vertical scrollbars:

- *Execution Output* — the `print()` values produced by stage 5, formatted as `value : Type`. The type label is determined at runtime by inspecting the Python type of the evaluated value; because the type checker guarantees correctness before execution, this always agrees with the statically inferred type;
- *Tokens* — the token stream from stage 1, one token per line with type, lexeme, and source position;

- *Parse Tree* — a grammar-oriented ASCII branch rendering of stage 2 using non-terminal labels such as `statement`, `assignment`, `expr`, `additive`, `term`, and `factor`;
- *AST* — the indented executable tree from stage 3;
- *Type Table* — the symbol table from stage 4 showing each name, kind, inferred type, initialisation state, and parameter list;
- *Errors* — populated only when a pipeline stage fails; shows the error message (e.g. [line 2, col 5] `TypeError: ...`). On a successful run this tab is cleared automatically.

**Error handling.** If any pipeline stage raises an exception, the error message is written exclusively to the *Errors* tab and the remaining tabs are cleared. No modal dialogs are used, so the user can edit and rerun without dismissing a popup.

**Keyboard shortcuts.** F5 and Ctrl+Enter are both bound to the *Run* action, allowing rapid edit-run cycles without reaching for the mouse.

### 6.6.1 Screenshots

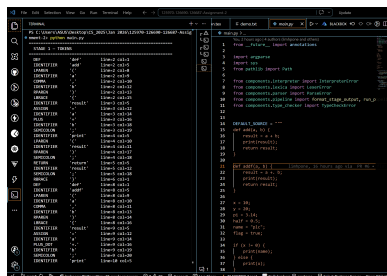
Figures 6.2 and 6.3 show the desktop workbench and CLI output.



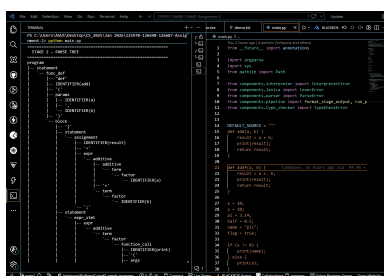
**Figure 6.2. Desktop workbench tabs.**

## 6.7 Command-line interface

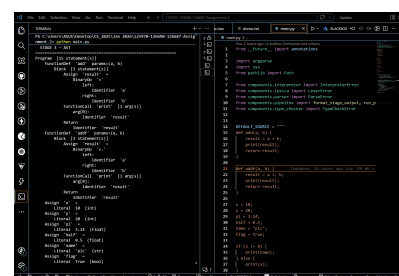
The CLI runner (`main.py`) accepts an optional script path; without one it runs the built-in sample and prints five labelled stages to `stdout`. Errors are reported cleanly to `stderr`: if the specified file does not exist, the runner prints `ERROR: File not found: <path>` and exits with code 1. If any pipeline stage raises a language error (`LexerError`, `ParseError`, `TypeCheckError`, or `InterpreterError`), the message is printed as `ERROR: [line X, col Y] <ErrorType>: <detail>` and the process exits with code 1. In both cases no Python traceback is shown to the user.



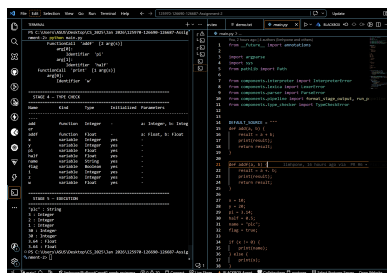
(a) Tokens.



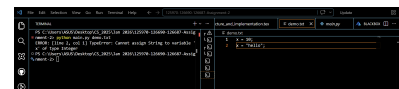
(b) Parse tree.



(c) AST.



(d) Types and output.



(e) CLI error.

Figure 6.3. CLI stages and error handling.

## CHAPTER 7

### TESTING AND VERIFICATION

#### 7.1 Manual Testing

Prior to the construction of the automated suite, each language feature was exercised manually by running the full pipeline and inspecting stage outputs. Table 7.1 summarises the categories covered.

Table 7.1. Representative manually tested categories

| Category      | Program Behaviour                                 | Expected Result                     |
|---------------|---------------------------------------------------|-------------------------------------|
| Arithmetic    | Evaluate integer and float expressions separately | Correct value and inferred type     |
| If-then-else  | Choose one branch from a Boolean condition        | Only matching branch runs           |
| While-loop    | Print during repeated execution                   | One output line per iteration       |
| Functions     | Infer parameters and return type                  | Valid call and correct return value |
| Type mismatch | Reassign incompatible type                        | Type error before runtime           |

##### 7.1.1 *Lexer and Symbol Table*

The lexer was verified manually by running the full pipeline and inspecting Stage 1 token output for programs that exercise every token category: integer and float literals, Boolean literals, string literals, the five reserved keywords, two-character comparison operators (`==`, `!=`), single-character operators, and delimiters. Identifier scanning with keyword fallback was confirmed by checking that `true` and `false` are emitted as `BOOL_LITERAL` tokens, the five reserved words (`if`, `else`, `while`, `def`, `return`) as their respective keyword token types, and ordinary names including `print` as `IDENTIFIER` tokens. For example, the source fragment `x = 10 ;` produces the token sequence:

```

1 IDENTIFIER  'x'      line=1 col=1
2 ASSIGN      '='      line=1 col=3
3 INT_LITERAL '10'     line=1 col=5
4 SEMICOLON   ';'      line=1 col=7

```

Error paths were also checked: supplying a bare `!` character raises a `LexerError` with a source position and a message suggesting `!=`, and an unterminated string literal raises a `LexerError` citing the opening line.

The symbol table was verified through the type checker and the Stage 3 output. Duplicate variable definitions were confirmed to raise a `SymbolTableError` before execution. Scope isolation was verified with nested blocks: a variable defined inside an `if` or `while` body is not visible in the outer scope after the block ends. The `format_table()` output was checked against the expected column layout (name, kind, type, initialisation status, parameters) for both variable and function symbols.

### ***7.1.2 Parser, Parse Tree, and AST***

The parser was verified manually by running the full pipeline and inspecting both the Stage 2 parse-tree output and the Stage 3 AST output. The following cases were checked:

- the parse-tree view contains grammar-oriented labels such as `program`, `statement`, `assignment`, and `expr`, confirming that the system now exposes a separate parse tree rather than only the executable AST,
- arithmetic expressions with mixed precedence (e.g. `x + y * 2`) produce a tree where multiplication is a deeper sub-tree than addition, confirming the `_expr/_additive/_term/_factor` precedence encoding,
- assignment versus expression statement dispatch: `x = 5`; produces an `Assign` node while `add(1,2)`; produces a `FunctionCall` node at the statement level, confirming the IDENTIFIER+ASSIGN lookahead,
- `if` with and without an `else` clause: the resulting `If` node has a non-null `else_block` only when the clause is present,
- function definitions store only parameter names in the `FunctionDef` node; the AST printer output was inspected to confirm no type annotations are present at this stage,
- string literals in `Literal` nodes have their surrounding double-quotes stripped; `print("hello")` prints `"hello"` : `String` at runtime (the interpreter re-adds the quotes when formatting string output),

- parse errors: missing semicolons, unmatched parentheses, and unexpected tokens all produce `[line X, col Y] ParseError`: messages with correct source positions.

### 7.1.3 Arithmetic Expressions

Arithmetic expressions were tested with integer-only and float-only programs to verify operator precedence and strict type separation. The expression `x + y * 2` confirmed that multiplication binds more tightly than addition, and mixing integer and float operands correctly produced a type error, confirming the no-overloading design. The test suite explicitly verifies that mixed-type arithmetic is rejected:

- `1 + 2` — valid Integer arithmetic.
- `2.0 +. 3.5` — valid Float arithmetic.
- `5.0 -. 1.5, 2.0 *. 3.0, 9.0 /. 4.0` — valid Float arithmetic for the other dot-suffixed operators.
- `1 + 2.5` — `TypeError`: operator `+` requires two Integer operands.
- `2 +. 3` — `TypeError`: operator `+.`  requires two Float operands.
- `10 / 0` and `10.0 /. 0.0` — `InterpreterError`: division by zero, raised at runtime.
- `1.0 +. 2.0 *. 3.0` — evaluates to `7.0` confirming `*.`  binds more tightly than `+.`

These rejection tests are not incidental — they are the primary evidence that the language satisfies the specification’s no-overloading requirement. In a promotion-based language, the expressions `1 + 2.5` and `2 +. 3` would silently succeed (returning `3.5 : Float` and `5.0 : Float` respectively). The fact that both are rejected as `TypeError`s demonstrates that each operator token has a single, fixed type signature and that no implicit type coercion occurs. The test suite contains 7 dedicated type-error tests for mixed-type and wrong-type operator usage (see Table 7.2, `TypeCheckerTests`), ensuring that the no-overloading guarantee is verified mechanically rather than by assertion alone.

### 7.1.4 Boolean Expressions

Boolean expressions were tested inside `if` and `while` conditions. The type checker correctly enforced that both operands of `==` and `!=` must be arithmetic; supplying a `Boolean` or `String` operand produced a source-positioned `TypeCheckError` before any execution.

### **7.1.5 Assignment Statements**

Assignments were tested for both first-use type inference and reassignment consistency. Assigning 5 to a variable and subsequently assigning "hello" to the same variable correctly produced a type error, confirming that the inferred type is fixed after first assignment.

### **7.1.6 If-Then-Else**

Conditional programs were constructed to verify that exactly one branch executes for a given condition and that neither branch is entered when the condition evaluates to the opposite truth value. Non-Boolean conditions were also supplied and confirmed to be rejected by the type checker before execution.

### **7.1.7 While-Loop**

While-loops were tested with an integer countdown from three to zero. The results confirmed that the loop body executes once per iteration and terminates when the condition becomes false, with `print()` producing one output line per pass rather than accumulating output until loop exit.

### **7.1.8 Functions**

Function tests covered value parameter passing, local scope execution, and return value propagation. The type checker correctly inferred parameter types from the first call site and enforced that signature for all subsequent calls to the same function.

### **7.1.9 Print**

The built-in `print()` function was tested with all supported runtime types. The output includes the printed value together with its type, such as `3.14 : Float` and `"plc" : String`.

### **7.1.10 Unary Minus**

The unary minus operator was verified for integer literals (`x = -5`), float literals (`y = -. 3.14`), negated float variables (`y = -. x`), negated integer variables (`y = -x`), and negation inside a larger expression (`x = 3 + -2`). In each case the type checker assigned the correct type and the interpreter produced the expected value. Note that float negation uses the `-.`  token, consistent with the Caml Light convention; using plain `-` on a `Float` operand is rejected

by the type checker.

## 7.2 Type Error Detection

The checker was verified with invalid programs involving incompatible assignment, invalid arithmetic operands, wrong function arguments, and non-Boolean conditions. In each case the system stopped before execution and reported a source-positioned type error. Representative error messages are shown below.

```
1 # Assignment type mismatch
2 [line 2, col 1] TypeError: Cannot assign String to variable 'x' of type
   Integer
3
4 # Non-Boolean if condition
5 [line 1, col 4] TypeError: If condition must have type Boolean
6
7 # Wrong argument type for function
8 [line 5, col 9] TypeError: Argument 1 of 'add' must be Integer, got
   String
```

## 7.3 Automated Test Suite

The automated regression suite is located in `tests/test_pipeline.py` and uses Python's built-in `unittest` framework. The tests are organised into five classes, each targeting a distinct compiler stage.

Table 7.2. Automated test coverage by class

| Test class       | Stage covered                                          | Tests     |
|------------------|--------------------------------------------------------|-----------|
| LexerTests       | Tokenisation ( <code>lexica.py</code> )                | 12        |
| SymbolTableTests | Scoped symbol table ( <code>symbol_table.py</code> )   | 7         |
| ParserTests      | Parsing, precedence, parse-tree/AST output             | 7         |
| TypeCheckerTests | Static type inference ( <code>type_checker.py</code> ) | 22        |
| IntegrationTests | Full pipeline execution                                | 27        |
| <b>Total</b>     |                                                        | <b>75</b> |

**LexerTests** verify that each token category (integer, float, Boolean, string, keywords, identifiers, two-character operators, and the four float-arithmetic operators `+`, `-`, `*`, `/`) is produced correctly, that float literals such as `3.14` are not confused with integer followed by a dot operator, that source line numbers are tracked across newlines, and that illegal characters and unterminated strings raise `LexerError`.

**ParserTests** verify structural correctness of parsing. The 7 tests cover missing-semicolon and unclosed-brace error reporting, integer operator precedence ( $*$  before  $+$ ), float operator precedence ( $*$ . before  $+$ .), left-associativity of arithmetic operators, parenthesised expressions overriding precedence, and availability of both the parse-tree and AST renderings in the pipeline result.

**SymbolTableTests** exercise the symbol table directly, without going through the full pipeline. They cover variable definition and lookup, duplicate definition raising `SymbolTableError` for both variables and functions, parent-scope lookup through `child_scope()`, undefined-symbol lookup error, and the `format_table()` output for both variable and function symbol rows.

**TypeCheckerTests** verify static type inference and error detection. The 22 tests cover integer and float arithmetic (all four dot-suffixed operators  $+$ .  $-$ .  $*$ .  $/$ . each verified to produce `Float`), integer floor division producing `Integer`, strict type separation (mixed `Integer/Float` rejected for both arithmetic and comparison operators, and integers used with a dot operator rejected), string and boolean type inference, assignment type mismatch, non-Boolean conditions, undefined variables, wrong argument count and type, and two tests for uncalled-function body checking: one confirms that a type error inside a never-called function body is detected, and another confirms that a valid uncalled function raises no error.

**IntegrationTests** run programs through all four pipeline stages and verify the final output. Cases include unary negation on all numeric types (including  $-$ . applied to a float variable), division by zero for both  $/$  and  $/$ . (each must raise `InterpreterError`), both branches of `if/else`, while-loop countdown, `print()` with every type, value-parameter isolation, nested function calls, comparison results assigned to variables and printed directly, variable reassignment, and scope isolation (variables declared inside an `if` block are not visible outside it).

The tests can be run with:

```
1 # macOS / Linux
2 python3 -m unittest discover -s tests -v
3
4 # Windows
5 .\venv\Scripts\python.exe -m unittest discover -s tests -v
```

All 75 tests pass.

## 7.4 Error Handling

The language system reports errors at three main stages:

- lexical errors for malformed characters or unterminated strings,
- parse errors for invalid syntax such as missing semicolons,
- type errors for programs that violate static typing rules.

Runtime errors are also reported with source positions whenever execution reaches an invalid state such as an undefined variable reference.

## CHAPTER 8

### CONCLUSION

This project delivered a complete, statically-typed interpreted language supporting Integer, Float, Boolean, and String types, with arithmetic, comparisons, assignments, if-then-else, while-loops, functions with value parameter passing, and a built-in `print()` function.

The system follows a four-stage pipeline: lexer, parser, type checker, and interpreter. Variable and function types are inferred without explicit declarations; programs that violate type rules are rejected before execution. For uncalled functions, the type checker performs a post-pass to infer parameter types from body usage, ensuring type errors are always caught. A central design decision — following the Caml Light convention of using separate operator tokens for integer and float arithmetic — ensures that the specification’s no-overloading requirement is satisfied in the strongest possible sense: every operator token carries exactly one type signature, and the compiler never inspects operand types to resolve operator meaning. This eliminates not only traditional overloading but also the implicit numeric promotion found in many mainstream languages, resulting in fully deterministic type inference where every expression’s type is derivable from the operator tokens and literal forms alone.

To support clearer correspondence between the formal grammar and accepted programs, the parser stage now exposes both a grammar-oriented parse-tree view and the executable AST used by later stages. The built-in `print()` operation is now implemented as a normal callable built-in rather than a dedicated statement form. The 75 automated tests across five classes — lexical analysis, symbol table, parsing, type checking, and full pipeline — confirm correct behaviour for all language features. Both a command-line runner and a desktop UI are provided.

The project demonstrates how lexical analysis, parsing, semantic analysis, type inference, and interpretation can be integrated into a coherent compiler pipeline with clear separation of responsibilities between stages. The implementation prioritised semantic clarity, deterministic type inference, and modular compiler structure over advanced language features. Future work could extend the language with recursive functions, richer data structures, and code generation.

## REFERENCES

- Alfred V. Aho, Monica S. Lam, Ravi Sethi, and Jeffrey D. Ullman. *Compilers: Principles, Techniques, and Tools*. Addison-Wesley, Boston, MA, 2nd edition, 2006.
- Robert Nystrom. *Crafting Interpreters*. Genever Benning, Seattle, WA, 2021.
- Python Software Foundation. Python documentation: dataclasses — data classes. <https://docs.python.org/3/library/dataclasses.html>, 2026a. Accessed: April 2026.
- Python Software Foundation. Python documentation: Tkinter — python interface to tcl/tk. <https://docs.python.org/3/library/tkinter.html>, 2026b. Accessed: April 2026.
- Python Software Foundation. Python documentation: unittest — unit testing framework. <https://docs.python.org/3/library/unittest.html>, 2026c. Accessed: April 2026.

## APPENDIX A: SOURCE CODE

The complete source code is available on GitHub:

<https://github.com/The-Token-Trio/125970-126690-126687-Assignment2-PLC>

The project can be run from the command line or the desktop UI:

```
1 # macOS / Linux
2 python3 main.py
3 python3 ui.py
4 python3 -m unittest discover -s tests -v
5
6 # Windows
7 .\venv\Scripts\python.exe main.py
8 .\venv\Scripts\python.exe ui.py
9 .\venv\Scripts\python.exe -m unittest discover -s tests -v
```

No external Python dependencies are required.

## APPENDIX B: TEAM MEMBERS AND CONTRIBUTIONS

| Name                                 | Student ID | Contributions                                                                                                                                                                                                      |
|--------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aye Khin Khin Hpone<br>(Yolanda Lim) | st125970   | Static typing rules; type checking; assignment, if, while, function, and <code>print()</code> execution; unary minus inference; pipeline integration ( <code>pipeline.py</code> ); automated test suite (75 tests) |
| Applegate T. Tun Oo                  | st126690   | Lexer implementation (scanning, tokenisation, line/column tracking, error reporting); token definitions; keywords, operators, identifiers, and literals; symbol table (variable/function/type storage)             |
| Win Htut Naing                       | st126687   | Arithmetic and boolean expressions; assignment, if-then-else, while-loop, function definition/call, and <code>print()</code> parsing; unary minus parsing                                                          |

### A.1 Entry Points

```
1 from __future__ import annotations
2
```

```

3 import argparse
4 import sys
5 from pathlib import Path
6
7 from components.interpreter import InterpreterError
8 from components.lexica import LexerError
9 from components.parser import ParseError
10 from components.pipeline import format_stage_output, run_pipeline
11 from components.type_checker import TypeCheckError
12
13
14 DEFAULT_SOURCE = """
15 def add(a, b) {
16     result = a + b;
17     print(result);
18     return result;
19 }
20
21 def addf(a, b) {
22     result = a +. b;
23     print(result);
24     return result;
25 }
26
27 x = 10;
28 y = 20;
29 pi = 3.14;
30 half = 0.5;
31 name = "plc";
32 flag = true;
33
34 if (x != 0) {
35     print(name);
36 } else {
37     print(x);
38 }
39
40 i = 3;
41 while (i != 0) {
42     print(i);
43     i = i - 1;
44 }
45
46 z = add(x, y);
47 print(z);
48
49 w = addf(pi, half);
50 print(w);
51 """
52
53
54 def run_source(source_code: str) -> None:
55     try:

```

```

56     result = run_pipeline(source_code)
57     print(format_stage_output(result))
58 except (LexerError, ParseError, TypeCheckError, InterpreterError)
59 as error:
60     print(f"ERROR: {error}", file=sys.stderr)
61     sys.exit(1)
62
63 def _parse_args() -> argparse.Namespace:
64     parser = argparse.ArgumentParser(description="Run the programming
65         language pipeline.")
66     parser.add_argument("script", nargs="?", help="Optional path to a
67         source file to run")
68     return parser.parse_args()
69
70 def main() -> None:
71     args = _parse_args()
72     if args.script:
73         path = Path(args.script)
74         if not path.exists():
75             print(f"ERROR: File not found: {args.script}", file=sys.
76                 stderr)
77             sys.exit(1)
78         source_code = path.read_text(encoding="utf-8")
79     else:
80         source_code = DEFAULT_SOURCE
81     run_source(source_code)
82
83 if __name__ == "__main__":
84     main()

```

Listing 8.1: main.py — command-line entry point

```

1 from __future__ import annotations
2
3 import tkinter as tk
4 from pathlib import Path
5 from tkinter import filedialog, ttk
6
7 from components.pipeline import format_tokens, run_pipeline
8
9
10 DEFAULT_SOURCE = """def add(a, b) {
11     result = a + b;
12     print(result);
13     return result;
14 }
15
16 def addf(a, b) {
17     result = a +. b;
18     print(result);

```

```

19     return result;
20 }
21
22 x = 10;
23 y = 20;
24 pi = 3.14;
25 half = 0.5;
26 name = \"plc\";
27 flag = true;
28
29 if (x != 0) {
30     print(name);
31 } else {
32     print(x);
33 }
34
35 i = 3;
36 while (i != 0) {
37     print(i);
38     i = i - 1;
39 }
40
41 z = add(x, y);
42 print(z);
43
44 w = addf(pi, half);
45 print(w);
46 """
47
48
49 class LanguageWorkbench:
50     def __init__(self) -> None:
51         self.root = tk.Tk()
52         self.root.title("125970-126690-126687-Ass2-PLC")
53         self.root.geometry("1200x760")
54         self.current_file: Path | None = None
55
56         self._build_layout()
57         self.source_text.insert("1.0", DEFAULT_SOURCE)
58         self.root.bind("<F5>", lambda _e: self._run_source())
59         self.root.bind("<Control-Return>", lambda _e: self._run_source
60             ())
61
62     def _build_layout(self) -> None:
63         toolbar = ttk.Frame(self.root, padding=10)
64         toolbar.pack(fill=tk.X)
65
66         ttk.Button(toolbar, text="Open Script", command=self.
67             _open_script).pack(side=tk.LEFT)
68         ttk.Button(toolbar, text="Run", command=self._run_source).pack(
69             side=tk.LEFT, padx=(8, 0))
70         ttk.Button(toolbar, text="Clear Output", command=self.
71             _clear_output).pack(side=tk.LEFT, padx=(8, 0))

```

```

68     self.status_var = tk.StringVar(value="Ready")
69     ttk.Label(toolbar, textvariable=self.status_var).pack(side=tk.
70         RIGHT)
71
72     body = ttk.Panedwindow(self.root, orient=tk.HORIZONTAL)
73     body.pack(fill=tk.BOTH, expand=True, padx=10, pady=(0, 10))
74
75     left_panel = ttk.Frame(body, padding=8)
76     right_panel = ttk.Frame(body, padding=8)
77     body.add(left_panel, weight=1)
78     body.add(right_panel, weight=1)
79
80     ttk.Label(left_panel, text="Input Script").pack(anchor=tk.W)
81     src_frame = ttk.Frame(left_panel)
82     src_frame.pack(fill=tk.BOTH, expand=True, pady=(6, 0))
83     self.source_text = tk.Text(src_frame, wrap=tk.NONE, font=(
84         "Consolas", 11))
85     src_vsb = ttk.Scrollbar(src_frame, orient=tk.VERTICAL, command=
86         self.source_text.yview)
87     src_hsb = ttk.Scrollbar(src_frame, orient=tk.HORIZONTAL,
88         command=self.source_text.xview)
89     self.source_text.configure(yscrollcommand=src_vsb.set,
90         xscrollcommand=src_hsb.set)
91     src_vsb.pack(side=tk.RIGHT, fill=tk.Y)
92     src_hsb.pack(side=tk.BOTTOM, fill=tk.X)
93     self.source_text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)
94
95     notebook = ttk.Notebook(right_panel)
96     notebook.pack(fill=tk.BOTH, expand=True)
97
98     self.output_text = self._make_output_tab(notebook, "Execution
99         Output")
100     self.tokens_text = self._make_output_tab(notebook, "Tokens")
101     self.parse_tree_text = self._make_output_tab(notebook, "Parse
102         Tree")
103     self.ast_text = self._make_output_tab(notebook, "AST")
104     self.types_text = self._make_output_tab(notebook, "Type Table")
105     self.error_text = self._make_output_tab(notebook, "Errors")
106
107     def _make_output_tab(self, notebook: ttk.Notebook, title: str) ->
108         tk.Text:
109         frame = ttk.Frame(notebook, padding=8)
110         notebook.add(frame, text=title)
111         text = tk.Text(frame, wrap=tk.NONE, font=("Consolas", 11),
112             state=tk.DISABLED)
113         vsb = ttk.Scrollbar(frame, orient=tk.VERTICAL, command=text.
114             yview)
115         hsb = ttk.Scrollbar(frame, orient=tk.HORIZONTAL, command=text.
116             xview)
117         text.configure(yscrollcommand=vsb.set, xscrollcommand=hsb.set)
118         vsb.pack(side=tk.RIGHT, fill=tk.Y)
119         hsb.pack(side=tk.BOTTOM, fill=tk.X)

```

```

110     text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)
111     return text
112
113     def _open_script(self) -> None:
114         file_path = filedialog.askopenfilename(
115             title="Open Script File",
116             filetypes=[("Text Files", "*.txt *.pl *.src"), ("All Files"
117                 , "*.*)"],
118         )
119         if not file_path:
120             return
121
122         path = Path(file_path)
123         self.current_file = path
124         self.source_text.delete("1.0", tk.END)
125         self.source_text.insert("1.0", path.read_text(encoding="utf-8")
126             )
127         self.status_var.set(f"Loaded {path.name}")
128
129     def _run_source(self) -> None:
130         source = self.source_text.get("1.0", tk.END)
131         try:
132             result = run_pipeline(source)
133         except Exception as error:
134             self._write_text(self.error_text, str(error))
135             self._write_text(self.output_text, "")
136             self._write_text(self.tokens_text, "")
137             self._write_text(self.parse_tree_text, "")
138             self._write_text(self.ast_text, "")
139             self._write_text(self.types_text, "")
140             self.status_var.set("Run failed")
141             return
142
143         self._write_text(self.output_text, "\n".join(result.outputs) if
144             result.outputs else "(no output)")
145         self._write_text(self.tokens_text, format_tokens(result.tokens)
146             )
147         self._write_text(self.parse_tree_text, result.parse_tree_text)
148         self._write_text(self.ast_text, result.ast_text)
149         self._write_text(self.types_text, result.checked_scope.
150             format_table())
151         self._write_text(self.error_text, "")
152
153         current_label = self.current_file.name if self.current_file is
154             not None else "editor contents"
155         self.status_var.set(f"Ran {current_label}")
156
157     def _clear_output(self) -> None:
158         self._write_text(self.output_text, "")
159         self._write_text(self.tokens_text, "")
160         self._write_text(self.parse_tree_text, "")
161         self._write_text(self.ast_text, "")
162         self._write_text(self.types_text, "")

```

```

157         self._write_text(self.error_text, "")
158         self.status_var.set("Output cleared")
159
160     @staticmethod
161     def _write_text(widget: tk.Text, content: str) -> None:
162         widget.config(state=tk.NORMAL)
163         widget.delete("1.0", tk.END)
164         widget.insert("1.0", content)
165         widget.config(state=tk.DISABLED)
166
167     def run(self) -> None:
168         self.root.mainloop()
169
170
171 if __name__ == "__main__":
172     LanguageWorkbench().run()

```

Listing 8.2: ui.py — Tkinter desktop workbench

## A.2 Core Components

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass
4 from enum import Enum, auto
5
6
7 class TokenType(Enum):
8     # Literals
9     INT_LITERAL = auto()
10    FLOAT_LITERAL = auto()
11    BOOL_LITERAL = auto()
12    STRING_LITERAL = auto()
13
14    # Identifier
15    IDENTIFIER = auto()
16
17    # Keywords
18    IF = auto()
19    ELSE = auto()
20    WHILE = auto()
21    DEF = auto()
22    RETURN = auto()
23
24    # Operators
25    PLUS = auto()
26    MINUS = auto()
27    STAR = auto()
28    SLASH = auto()
29    PLUS_DOT = auto()    # +. (float addition)
30    MINUS_DOT = auto()   # -. (float subtraction)
31    STAR_DOT = auto()    # *. (float multiplication)

```

```

32 SLASH_DOT = auto()    # /.  (float division)
33 EQUAL_EQUAL = auto()
34 BANG_EQUAL = auto()
35 ASSIGN = auto()
36
37 # Delimiters
38 LPAREN = auto()
39 RPAREN = auto()
40 LBRACE = auto()
41 RBRACE = auto()
42 COMMA = auto()
43 SEMICOLON = auto()
44
45 EOF = auto()
46
47
48 @dataclass(frozen=True)
49 class Token:
50     token_type: TokenType
51     lexeme: str
52     line: int
53     column: int

```

Listing 8.3: components/tokens.py — token type enumeration and Token dataclass

```

1 from __future__ import annotations
2
3 from typing import Iterator
4
5 from components.tokens import Token, TokenType
6
7
8 KEYWORDS: dict[str, TokenType] = {
9     "if": TokenType.IF,
10    "else": TokenType.ELSE,
11    "while": TokenType.WHILE,
12    "def": TokenType.DEF,
13    "return": TokenType.RETURN,
14    "true": TokenType.BOOL_LITERAL,
15    "false": TokenType.BOOL_LITERAL,
16 }
17
18
19 # Arithmetic operators that may have a trailing '.' for the float
   variant
20 _ARITH_DOT: dict[str, tuple[TokenType, TokenType]] = {
21     "+": (TokenType.PLUS, TokenType.PLUS_DOT),
22     "-": (TokenType.MINUS, TokenType.MINUS_DOT),
23     "*": (TokenType.STAR, TokenType.STAR_DOT),
24     "/": (TokenType.SLASH, TokenType.SLASH_DOT),
25 }
26
27 SINGLE_CHAR_TOKENS: dict[str, TokenType] = {

```

```

28     "=": TokenType.ASSIGN,
29     "(": TokenType.LPAREN,
30     ")": TokenType.RPAREN,
31     "{": TokenType.LBRACE,
32     "}": TokenType.RBRACE,
33     ",": TokenType.COMMA,
34     ";": TokenType.SEMICOLON,
35 }
36
37
38 class LexerError(Exception):
39     pass
40
41
42 class Lexer:
43     def __init__(self, source: str) -> None:
44         self.source = source
45         self.start = 0
46         self.current = 0
47         self.line = 1
48         self.column = 1
49         self.token_column = 1
50
51     def tokenize(self) -> list[Token]:
52         tokens: list[Token] = []
53         while not self._is_at_end():
54             self.start = self.current
55             self.token_column = self.column
56             token = self._scan_token()
57             if token is not None:
58                 tokens.append(token)
59         tokens.append(Token(TokenType.EOF, "", self.line, self.column))
60         return tokens
61
62     def _scan_token(self) -> Token | None:
63         char = self._advance()
64
65         # Whitespace handling
66         if char in {" ", "\t", "\r"}:
67             return None
68         if char == "\n":
69             self.line += 1
70             self.column = 1
71             return None
72
73         # Two-char operators
74         if char == "=":
75             if self._match("=="):
76                 return self._make_token(TokenType.EQUAL_EQUAL)
77             return self._make_token(TokenType.ASSIGN)
78         if char == "!":
79             if self._match("!="):
80                 return self._make_token(TokenType.BANG_EQUAL)

```

```

81         raise LexerError(self._error_message("Unexpected '!'. Did
82             you mean '!= '?'"))
83
84     # Arithmetic operators: + - * / (with optional trailing '.'
85     # for float variant)
86     arith = _ARITH_DOT.get(char)
87     if arith is not None:
88         int_tok, float_tok = arith
89         # Unary-minus lookahead: '-' followed by a digit is NOT
90         # '-' even if
91         # next char is '.' -- we still need to check char after '.'
92         # is not digit
93         # (to avoid consuming the '.' of a float literal like 3.14)
94         # Rule: X. is a float operator only when the char after '.'
95         # is NOT a digit.
96         if self._peek() == "." and not self._peek_next().isdigit():
97             self._advance() # consume '.'
98             return self._make_token(float_tok)
99         return self._make_token(int_tok)
100
101     # Single-char operators/delimiters
102     single_char_token = SINGLE_CHAR_TOKENS.get(char)
103     if single_char_token is not None:
104         return self._make_token(single_char_token)
105
106     if char == '"':
107         return self._string()
108
109     if char.isdigit():
110         return self._number()
111
112     if self._is_identifier_start(char):
113         return self._identifier()
114
115     raise LexerError(self._error_message(f"Unexpected character: {
116         char!r}"))
117
118 def _string(self) -> Token:
119     while self._peek() != '"' and not self._is_at_end():
120         if self._peek() == "\n":
121             self.line += 1
122             self.column = 1
123             self._advance()
124
125     if self._is_at_end():
126         raise LexerError(self._error_message("Unterminated string
127             literal"))
128
129     # Closing quote
130     self._advance()
131     return self._make_token(TokenType.STRING_LITERAL)

```

```

126 def _number(self) -> Token:
127     while self._peek().isdigit():
128         self._advance()
129
130     is_float = False
131     if self._peek() == "." and self._peek_next().isdigit():
132         is_float = True
133         self._advance()
134         while self._peek().isdigit():
135             self._advance()
136
137     if is_float:
138         return self._make_token(TokenType.FLOAT_LITERAL)
139     return self._make_token(TokenType.INT_LITERAL)
140
141 def _identifier(self) -> Token:
142     while self._is_identifier_part(self._peek()):
143         self._advance()
144
145     lexeme = self._current_lexeme()
146     token_type = KEYWORDS.get(lexeme, TokenType.IDENTIFIER)
147     return self._make_token(token_type)
148
149 def _current_lexeme(self) -> str:
150     return self.source[self.start : self.current]
151
152 def _make_token(self, token_type: TokenType) -> Token:
153     return Token(token_type, self._current_lexeme(), self.line,
154                 self.token_column)
155
156 def _advance(self) -> str:
157     char = self.source[self.current]
158     self.current += 1
159     self.column += 1
160     return char
161
162 def _match(self, expected: str) -> bool:
163     if self._is_at_end():
164         return False
165     if self.source[self.current] != expected:
166         return False
167
168     self.current += 1
169     self.column += 1
170     return True
171
172 def _peek(self) -> str:
173     if self._is_at_end():
174         return "\0"
175     return self.source[self.current]
176
177 def _peek_next(self) -> str:
178     if self.current + 1 >= len(self.source):

```

```

178         return "\0"
179         return self.source[self.current + 1]
180
181     def _is_at_end(self) -> bool:
182         return self.current >= len(self.source)
183
184     @staticmethod
185     def _is_identifier_start(char: str) -> bool:
186         return char.isalpha() or char == "_"
187
188     @staticmethod
189     def _is_identifier_part(char: str) -> bool:
190         return char.isalnum() or char == "_"
191
192     def _error_message(self, message: str) -> str:
193         return f"[line {self.line}, col {self.token_column}] {message}"
194
195
196 def iter_tokens(source: str) -> Iterator[Token]:
197     yield from Lexer(source).tokenize()

```

Listing 8.4: components/lexica.py — lexical analyser

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass, field
4 from typing import Optional
5
6
7 #
8     -----
9
10 # Base
11 #
12     -----
13
14 @dataclass
15 class ASTNode:
16     """Base class for every AST node. Carries source location for
17     diagnostics."""
18     line: int = field(default=0, kw_only=True)
19     column: int = field(default=0, kw_only=True)
20
21 #
22     -----
23
24 # Expressions
25 #
26     -----

```

```

22 @dataclass
23 class Literal(ASTNode):
24     """An integer, float, boolean, or string literal.
25
26     Examples:  42    3.14    true    "hello"
27     """
28     value: int | float | bool | str
29
30
31 @dataclass
32 class Identifier(ASTNode):
33     """A variable or function name used as an expression.
34
35     Example:  x
36     """
37     name: str
38
39
40 @dataclass
41 class BinaryOp(ASTNode):
42     """An arithmetic or comparison binary operation.
43
44     op is one of: '+' '-' '*' '/' '+.' '-.' '*.' '/.' '==' '!='
45     Example:  a + b    x == 0    a +. b
46     """
47     op: str
48     left: ASTNode
49     right: ASTNode
50
51
52 @dataclass
53 class FunctionCall(ASTNode):
54     """A call to a user-defined function.
55
56     Example:  add(1, 2)
57     """
58     name: str
59     args: list[ASTNode] = field(default_factory=list)
60
61
62 #
63 # -----
64 #
65
66 # Statements
67 # -----
68
69
70 @dataclass
71 class Assign(ASTNode):
72     """Variable assignment (first use infers type; later uses must
73     match).

```

```

70     Example:  x = 10;
71     """
72     name: str
73     value: ASTNode
74
75
76 @dataclass
77 class Return(ASTNode):
78     """Return statement inside a function body.
79
80     Example:  return result;
81     """
82     expr: ASTNode
83
84
85 @dataclass
86 class Block(ASTNode):
87     """A brace-enclosed sequence of statements.
88
89     Example:  { x = 1; print(x); }
90     """
91     statements: list[ASTNode] = field(default_factory=list)
92
93
94 @dataclass
95 class If(ASTNode):
96     """If / optional else control flow.
97
98     Example:  if (x != 0) { print(x); } else { print(0); }
99     """
100     condition: ASTNode
101     then_block: Block
102     else_block: Optional[Block]
103
104
105 @dataclass
106 class While(ASTNode):
107     """While loop.
108
109     Example:  while (x != 0) { x = x - 1; }
110     """
111     condition: ASTNode
112     body: Block
113
114
115 @dataclass
116 class FunctionDef(ASTNode):
117     """Function definition.
118
119     Example:  def add(a, b) { return a + b; }
120
121     Note: parameter types are unknown at parse time -- Member 3's type
        checker

```

```

122     will infer them from usage. We store only the parameter names here.
123     """
124     name: str
125     params: list[str]
126     body: Block
127
128
129     #
130     -----
131
132     # Top-level
133     #
134     -----
135
136     @dataclass
137     class Program(ASTNode):
138         """Root node: the entire source file as a list of statements."""
139         statements: list[ASTNode] = field(default_factory=list)

```

Listing 8.5: components/ast\_nodes.py — AST node definitions

```

1 from __future__ import annotations
2
3 from typing import Optional
4
5 from components.tokens import Token, TokenType
6 from components.ast_nodes import (
7     ASTNode,
8     Program,
9     Block,
10    Assign,
11    If,
12    While,
13    FunctionDef,
14    FunctionCall,
15    Return,
16    BinaryOp,
17    Literal,
18    Identifier,
19 )
20
21
22 #
23 -----
24
25 # Parser error
26 #
27 -----
28
29 class ParseError(Exception):
30     pass

```

```

28
29
30 #
    -----
31 # Parser
32 #
    -----
33
34 class Parser:
35     """Recursive-descent parser.
36
37     Usage:
38         tokens = Lexer(source).tokenize()
39         tree   = Parser(tokens).parse()    # returns Program node
40     """
41
42     def __init__(self, tokens: list[Token]) -> None:
43         self._tokens = tokens
44         self._pos = 0
45
46     #
47     -----
48     # Public entry point
49     #
50     -----
51
52     def parse(self) -> Program:
53         """Parse the full token stream and return the root Program node
54         ."""
55         line, col = self._peek().line, self._peek().column
56         statements: list[ASTNode] = []
57         while not self._is_at_end():
58             statements.append(self._statement())
59         return Program(statements=statements, line=line, column=col)
60
61     #
62     -----
63
64     # Statements
65     #
66     -----
67
68     def _statement(self) -> ASTNode:
69         """Dispatch to the correct statement parser based on the
70         current token."""
71         tok = self._peek()
72
73         if tok.token_type == TokenType.DEF:

```

```

67         return self._func_def()
68
69     if tok.token_type == TokenType.IF:
70         return self._if_stmt()
71
72     if tok.token_type == TokenType.WHILE:
73         return self._while_stmt()
74
75     if tok.token_type == TokenType.RETURN:
76         return self._return_stmt()
77
78     # assignment: IDENTIFIER "=" expr ";"
79     if (
80         tok.token_type == TokenType.IDENTIFIER
81         and self._peek_next().token_type == TokenType.ASSIGN
82     ):
83         return self._assignment()
84
85     # fallback: bare expression statement expr ";"
86     node = self._expr()
87     self._expect(TokenType.SEMICOLON, "Expected ';' after
88         expression")
89     return node
90
91     def _assignment(self) -> Assign:
92         """IDENTIFIER '=' expr ';'"""
93         name_tok = self._advance() # IDENTIFIER
94         self._advance() # '='
95         value = self._expr()
96         self._expect(TokenType.SEMICOLON, "Expected ';' after
97             assignment")
98         return Assign(name=name_tok.lexeme, value=value,
99             line=name_tok.line, column=name_tok.column)
100
101     def _if_stmt(self) -> If:
102         """'if' '(' expr ')' block ( 'else' block )?"""
103         tok = self._advance() # 'if'
104         self._expect(TokenType.LPAREN, "Expected '(' after 'if'")
105         condition = self._expr()
106         self._expect(TokenType.RPAREN, "Expected ')' after if condition
107             ")
108         then_block = self._block()
109         else_block: Optional[Block] = None
110         if self._check(TokenType.ELSE):
111             self._advance() # 'else'
112             else_block = self._block()
113         return If(condition=condition, then_block=then_block,
114             else_block=else_block,
115             line=tok.line, column=tok.column)
116
117     def _while_stmt(self) -> While:

```

```

114     """'while' '(' expr ')' block"""
115     tok = self._advance() # '
116         while'
117     self._expect(TokenType.LPAREN, "Expected '(' after 'while'")
118     condition = self._expr()
119     self._expect(TokenType.RPAREN, "Expected ')' after while
120         condition")
121     body = self._block()
122     return While(condition=condition, body=body,
123         line=tok.line, column=tok.column)
124
125 def _func_def(self) -> FunctionDef:
126     """'def' IDENTIFIER '(' params? ')' block"""
127     tok = self._advance() # 'def'
128     name_tok = self._expect(TokenType.IDENTIFIER, "Expected
129         function name after 'def'")
130     self._expect(TokenType.LPAREN, "Expected '(' after function
131         name")
132     params = self._params()
133     self._expect(TokenType.RPAREN, "Expected ')' after parameters")
134     body = self._block()
135     return FunctionDef(name=name_tok.lexeme, params=params, body=
136         body,
137         line=tok.line, column=tok.column)
138
139 def _return_stmt(self) -> Return:
140     """'return' expr ';'"""
141     tok = self._advance() # '
142         return'
143     expr = self._expr()
144     self._expect(TokenType.SEMICOLON, "Expected ';' after return
145         value")
146     return Return(expr=expr, line=tok.line, column=tok.column)
147
148 def _block(self) -> Block:
149     """'{' statement* '}'"""
150     tok = self._expect(TokenType.LBRACE, "Expected '{'")
151     statements: list[ASTNode] = []
152     while not self._check(TokenType.RBRACE) and not self._is_at_end
153         ():
154         statements.append(self._statement())
155     self._expect(TokenType.RBRACE, "Expected '}' to close block")
156     return Block(statements=statements, line=tok.line, column=tok.
157         column)
158
159 #
160
161 # Parameters (function definition)
162 #

```

```

154 def _params(self) -> list[str]:
155     """IDENTIFIER (',' IDENTIFIER)* -- returns list of param names.
        """
156     params: list[str] = []
157     if self._check(TokenType.IDENTIFIER):
158         params.append(self._advance().lexeme)
159         while self._check(TokenType.COMMA):
160             self._advance() # ','
161             tok = self._expect(TokenType.IDENTIFIER, "Expected
        parameter name after ','")
162             params.append(tok.lexeme)
163     return params
164
165     #
        -----
166     # Expressions (comparison is lowest precedence, then additive,
        then term)
167     #
        -----
168
169 def _expr(self) -> ASTNode:
170     """additive (('==' | '!=') additive)? -- non-associative
        comparison"""
171     node = self._additive()
172     if self._check(TokenType.EQUAL_EQUAL) or self._check(TokenType.
        BANG_EQUAL):
173         op_tok = self._advance()
174         right = self._additive()
175         node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
176             line=op_tok.line, column=op_tok.column)
177     return node
178
179 def _additive(self) -> ASTNode:
180     """term (('+' | '-' | '+.' | '-.') term)*"""
181     node = self._term()
182     while self._check(TokenType.PLUS) or self._check(TokenType.
        MINUS) \
183         or self._check(TokenType.PLUS_DOT) or self._check(
        TokenType.MINUS_DOT):
184         op_tok = self._advance()
185         right = self._term()
186         node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
187             line=op_tok.line, column=op_tok.column)
188     return node
189
190 def _term(self) -> ASTNode:
191     """factor (('*' | '/' | '*.' | '/.') factor)*"""
192     node = self._factor()
193     while self._check(TokenType.STAR) or self._check(TokenType.
        SLASH) \
194         or self._check(TokenType.STAR_DOT) or self._check(

```

```

                TokenType.SLASH_DOT):
195         op_tok = self._advance()
196         right = self._factor()
197         node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
198                        line=op_tok.line, column=op_tok.column)
199     return node
200
201     def _factor(self) -> ASTNode:
202         """
203         '-' factor (unary negation, desugared to 0 - factor)
204         | INT_LITERAL | FLOAT_LITERAL | STRING_LITERAL | BOOL_LITERAL
205         | IDENTIFIER ( '(' args? ')' )?
206         | '(' expr ')'
207         """
208         tok = self._peek()
209
210         # Unary minus '-' factor (integer negation, desugared to 0 -
211         # factor)
212         if tok.token_type == TokenType.MINUS:
213             op_tok = self._advance()
214             operand = self._factor()
215             zero = Literal(value=0, line=op_tok.line, column=op_tok.
216                           column)
217             return BinaryOp(op="-", left=zero, right=operand,
218                            line=op_tok.line, column=op_tok.column)
219
220         # Unary minus-dot '-.' factor (float negation, desugared to
221         # 0.0 -. factor)
222         if tok.token_type == TokenType.MINUS_DOT:
223             op_tok = self._advance()
224             operand = self._factor()
225             zero_f = Literal(value=0.0, line=op_tok.line, column=op_tok.
226                             column)
227             return BinaryOp(op="-.", left=zero_f, right=operand,
228                            line=op_tok.line, column=op_tok.column)
229
230         # Integer literal
231         if tok.token_type == TokenType.INT_LITERAL:
232             self._advance()
233             return Literal(value=int(tok.lexeme), line=tok.line, column
234                           =tok.column)
235
236         # Float literal
237         if tok.token_type == TokenType.FLOAT_LITERAL:
238             self._advance()
239             return Literal(value=float(tok.lexeme), line=tok.line,
240                           column=tok.column)
241
242         # Boolean literal
243         if tok.token_type == TokenType.BOOL_LITERAL:
244             self._advance()
245             return Literal(value=(tok.lexeme == "true"), line=tok.line,
246                           column=tok.column)

```

```

240
241 # String literal (keep the surrounding quotes stripped)
242 if tok.token_type == TokenType.STRING_LITERAL:
243     self._advance()
244     return Literal(value=tok.lexeme[1:-1], line=tok.line,
245                    column=tok.column)
246
247 # Identifier -- could be a plain variable or a function call
248 if tok.token_type == TokenType.IDENTIFIER:
249     self._advance()
250     if self._check(TokenType.LPAREN):
251         # function call
252         self._advance() # '(',
253         args = self._args()
254         self._expect(TokenType.RPAREN, "Expected ')' after
255                        function arguments")
256         return FunctionCall(name=tok.lexeme, args=args,
257                             line=tok.line, column=tok.column)
258     return Identifier(name=tok.lexeme, line=tok.line, column=
259                       tok.column)
260
261 # Grouped expression '(' expr ')'
262 if tok.token_type == TokenType.LPAREN:
263     self._advance() # '(',
264     node = self._expr()
265     self._expect(TokenType.RPAREN, "Expected ')' to close
266                        grouped expression")
267     return node
268
269 raise ParseError(self._error_at(tok, f"Unexpected token: {tok.
270                                lexeme!r}"))
271
272 #
273 -----
274
275 # Arguments (function call)
276 #
277 -----
278
279 def _args(self) -> list[ASTNode]:
280     """expr (',' expr)*"""
281     args: list[ASTNode] = []
282     if not self._check(TokenType.RPAREN):
283         args.append(self._expr())
284         while self._check(TokenType.COMMA):
285             self._advance() # ',',
286             args.append(self._expr())
287     return args
288
289 #
290 -----

```

```

282     # Token navigation helpers
283     #
284     -----
285     def _peek(self) -> Token:
286         return self._tokens[self._pos]
287
288     def _peek_next(self) -> Token:
289         if self._pos + 1 < len(self._tokens):
290             return self._tokens[self._pos + 1]
291         return self._tokens[-1] # EOF
292
293     def _advance(self) -> Token:
294         tok = self._tokens[self._pos]
295         if not self._is_at_end():
296             self._pos += 1
297         return tok
298
299     def _check(self, token_type: TokenType) -> bool:
300         return self._peek().token_type == token_type
301
302     def _is_at_end(self) -> bool:
303         return self._peek().token_type == TokenType.EOF
304
305     def _expect(self, token_type: TokenType, message: str) -> Token:
306         if self._check(token_type):
307             return self._advance()
308         raise ParseError(self._error_at(self._peek(), message))
309
310     def _error_at(self, tok: Token, message: str) -> str:
311         return f"[line {tok.line}, col {tok.column}] ParseError: {
            message}"

```

Listing 8.6: components/parser.py — recursive-descent parser

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass, field
4
5 from components.ast_nodes import (
6     ASTNode,
7     Assign,
8     BinaryOp,
9     Block,
10    FunctionCall,
11    FunctionDef,
12    Identifier,
13    If,
14    Literal,
15    Program,
16    Return,
17    While,

```

```

18 )
19
20
21 @dataclass
22 class ParseTreeNode:
23     label: str
24     children: list["ParseTreeNode"] = field(default_factory=list)
25
26
27 class ParseTreePrinter:
28     """Renders a textbook-style concrete parse tree using ASCII
29     branches."""
30
31     def print(self, node: Program) -> str:
32         root = self._program_node(node)
33         lines: list[str] = []
34         self._render(root, lines, prefix="", is_last=True, is_root=True)
35         return "\n".join(lines)
36
37     def _render(
38         self,
39         node: ParseTreeNode,
40         lines: list[str],
41         prefix: str,
42         is_last: bool,
43         is_root: bool = False,
44     ) -> None:
45         if is_root:
46             lines.append(node.label)
47         else:
48             branch = "'-- " if is_last else "|-- "
49             lines.append(f"{prefix}{branch}{node.label}")
50
51         child_prefix = prefix + ("    " if is_last else "|    ")
52         for index, child in enumerate(node.children):
53             self._render(
54                 child,
55                 lines,
56                 prefix=child_prefix if not is_root else "",
57                 is_last=index == len(node.children) - 1,
58             )
59
60     def _program_node(self, node: Program) -> ParseTreeNode:
61         children = [self._statement_node(statement) for statement in
62                     node.statements]
63         children.append(ParseTreeNode("EOF"))
64         return ParseTreeNode("program", children)
65
66     def _statement_node(self, node: ASTNode) -> ParseTreeNode:
67         if isinstance(node, Assign):
68             return ParseTreeNode("statement", [self._assignment_node(
69                 node)])

```

```

67     if isinstance(node, If):
68         return ParseTreeNode("statement", [self._if_node(node)])
69     if isinstance(node, While):
70         return ParseTreeNode("statement", [self._while_node(node)])
71     if isinstance(node, FunctionDef):
72         return ParseTreeNode("statement", [self._function_def_node(
73             node)])
74     if isinstance(node, Return):
75         return ParseTreeNode("statement", [self._return_node(node)
76             ])
77     return ParseTreeNode(
78         "statement",
79         [
80             ParseTreeNode(
81                 "expr_stmt",
82                 [self._expr_node(node), ParseTreeNode("';'")],
83             )
84         ],
85     )
86
87     def _assignment_node(self, node: Assign) -> ParseTreeNode:
88         return ParseTreeNode(
89             "assignment",
90             [
91                 ParseTreeNode(f"IDENTIFIER({node.name})", ),
92                 ParseTreeNode("'='"),
93                 self._expr_node(node.value),
94                 ParseTreeNode("';'"),
95             ],
96         )
97
98     def _if_node(self, node: If) -> ParseTreeNode:
99         children = [
100             ParseTreeNode("'if'"),
101             ParseTreeNode("'('"),
102             self._expr_node(node.condition),
103             ParseTreeNode("')'"),
104             self._block_node(node.then_block),
105         ]
106         if node.else_block is not None:
107             children.extend([ParseTreeNode("'else'"), self._block_node(
108                 node.else_block)])
109         return ParseTreeNode("if_stmt", children)
110
111     def _while_node(self, node: While) -> ParseTreeNode:
112         return ParseTreeNode(
113             "while_stmt",
114             [
115                 ParseTreeNode("'while'"),
116                 ParseTreeNode("'('"),
117                 self._expr_node(node.condition),
118                 ParseTreeNode("')'"),
119                 self._block_node(node.body),

```

```

117         ],
118     )
119
120     def _function_def_node(self, node: FunctionDef) -> ParseTreeNode:
121         return ParseTreeNode(
122             "func_def",
123             [
124                 ParseTreeNode("'def'"),
125                 ParseTreeNode(f"IDENTIFIER({node.name})"),
126                 ParseTreeNode("'('"),
127                 self._params_node(node.params),
128                 ParseTreeNode("')'"),
129                 self._block_node(node.body),
130             ],
131         )
132
133     def _params_node(self, params: list[str]) -> ParseTreeNode:
134         if not params:
135             return ParseTreeNode("params", [ParseTreeNode("epsilon")])
136
137         children: list[ParseTreeNode] = []
138         for index, param in enumerate(params):
139             children.append(ParseTreeNode(f"IDENTIFIER({param})"))
140             if index < len(params) - 1:
141                 children.append(ParseTreeNode("','"))
142         return ParseTreeNode("params", children)
143
144     def _return_node(self, node: Return) -> ParseTreeNode:
145         return ParseTreeNode(
146             "return_stmt",
147             [
148                 ParseTreeNode("'return'"),
149                 self._expr_node(node.expr),
150                 ParseTreeNode("';'"),
151             ],
152         )
153
154     def _block_node(self, node: Block) -> ParseTreeNode:
155         children = [ParseTreeNode("'{'")]
156         children.extend(self._statement_node(statement) for statement
157                        in node.statements)
158         children.append(ParseTreeNode("'}'"))
159         return ParseTreeNode("block", children)
160
161     def _expr_node(self, node: ASTNode) -> ParseTreeNode:
162         if isinstance(node, BinaryOp) and node.op in {"==", "!="}:
163             child = ParseTreeNode(
164                 "comparison",
165                 [
166                     self._additive_node(node.left),
167                     ParseTreeNode(f"{'{node.op}'")
168                     self._additive_node(node.right),
169                 ],
170             )

```

```

169         )
170         return ParseTreeNode("expr", [child])
171     return ParseTreeNode("expr", [self._additive_node(node)])
172
173     def _additive_node(self, node: ASTNode) -> ParseTreeNode:
174         if isinstance(node, BinaryOp) and node.op in {"+", "-", "+.", "-."}:
175             return ParseTreeNode(
176                 "additive",
177                 [
178                     self._additive_node(node.left),
179                     ParseTreeNode(f" '{node.op}'"),
180                     self._term_node(node.right),
181                 ],
182             )
183         return ParseTreeNode("additive", [self._term_node(node)])
184
185     def _term_node(self, node: ASTNode) -> ParseTreeNode:
186         if isinstance(node, BinaryOp) and node.op in {"*", "/", "*.", "/."}:
187             return ParseTreeNode(
188                 "term",
189                 [
190                     self._term_node(node.left),
191                     ParseTreeNode(f" '{node.op}'"),
192                     self._factor_node(node.right),
193                 ],
194             )
195         return ParseTreeNode("term", [self._factor_node(node)])
196
197     def _factor_node(self, node: ASTNode) -> ParseTreeNode:
198         if self._is_integer_unary_minus(node):
199             binary = node
200             assert isinstance(binary, BinaryOp)
201             return ParseTreeNode("factor", [ParseTreeNode("'-')", self._factor_node(binary.right)])
202         if self._is_float_unary_minus(node):
203             binary = node
204             assert isinstance(binary, BinaryOp)
205             return ParseTreeNode("factor", [ParseTreeNode("'-.')", self._factor_node(binary.right)])
206         if isinstance(node, Literal):
207             return ParseTreeNode("factor", [ParseTreeNode(self._literal_label(node.value))])
208         if isinstance(node, Identifier):
209             return ParseTreeNode("factor", [ParseTreeNode(f"IDENTIFIER ({node.name})")])
210         if isinstance(node, FunctionCall):
211             return ParseTreeNode("factor", [self._function_call_node(node)])
212         return ParseTreeNode(
213             "factor",
214             [ParseTreeNode("'('"), self._expr_node(node), ParseTreeNode("')'")]

```

```

        (",",')'],
215     )
216
217 def _function_call_node(self, node: FunctionCall) -> ParseTreeNode:
218     return ParseTreeNode(
219         "function_call",
220         [
221             ParseTreeNode(f"IDENTIFIER({node.name})",),
222             ParseTreeNode("','"),
223             self._args_node(node.args),
224             ParseTreeNode("','"),
225         ],
226     )
227
228 def _args_node(self, args: list[ASTNode]) -> ParseTreeNode:
229     if not args:
230         return ParseTreeNode("args", [ParseTreeNode("epsilon")])
231
232     children: list[ParseTreeNode] = []
233     for index, arg in enumerate(args):
234         children.append(self._expr_node(arg))
235         if index < len(args) - 1:
236             children.append(ParseTreeNode("','"))
237     return ParseTreeNode("args", children)
238
239 @staticmethod
240 def _is_integer_unary_minus(node: ASTNode) -> bool:
241     return (
242         isinstance(node, BinaryOp)
243         and node.op == "-"
244         and isinstance(node.left, Literal)
245         and node.left.value == 0
246     )
247
248 @staticmethod
249 def _is_float_unary_minus(node: ASTNode) -> bool:
250     return (
251         isinstance(node, BinaryOp)
252         and node.op == "-."
253         and isinstance(node.left, Literal)
254         and node.left.value == 0.0
255     )
256
257 @staticmethod
258 def _literal_label(value: object) -> str:
259     if isinstance(value, bool):
260         return f"BOOL_LITERAL({'true' if value else 'false'})"
261     if isinstance(value, int):
262         return f"INT_LITERAL({value})"
263     if isinstance(value, float):
264         return f"FLOAT_LITERAL({value})"
265     if isinstance(value, str):
266         return f"STRING_LITERAL('{value}'),

```

267

```
return f"LITERAL({value!r})"
```

Listing 8.7: components/parse\_tree\_printer.py — parse-tree renderer

```

1 from __future__ import annotations
2
3 from components.ast_nodes import (
4     ASTNode,
5     Program,
6     Block,
7     Assign,
8     If,
9     While,
10    FunctionDef,
11    FunctionCall,
12    Return,
13    BinaryOp,
14    Literal,
15    Identifier,
16 )
17
18
19 class ASTPrinter:
20     """Walks an AST and returns a human-readable, indented string.
21
22     Usage:
23         tree = Parser(tokens).parse()
24         print(ASTPrinter().print(tree))
25
26     Output style -- resembles the original source but annotated with
27         node types,
28     making it easy to verify the tree structure during development and
29     in reports.
30     """
31
32     INDENT = "    " # 4 spaces per level
33
34     def print(self, node: ASTNode) -> str:
35         lines: list[str] = []
36         self._visit(node, lines, level=0)
37         return "\n".join(lines)
38
39     # -----
40
41     # Dispatcher
42     # -----
43
44     def _visit(self, node: ASTNode, lines: list[str], level: int) -> None:
45         method_name = "_visit_" + type(node).__name__

```

```

43     visitor = getattr(self, method_name, self._visit_unknown)
44     visitor(node, lines, level)
45
46     def _pad(self, level: int) -> str:
47         return self.INDENT * level
48
49     #
50     # Node visitors
51     #
52
53     def _visit_Program(self, node: Program, lines: list[str], level:
54         int) -> None:
55         lines.append(f"{self._pad(level)}Program [{len(node.statements
56             )} statement(s)]")
57         for stmt in node.statements:
58             self._visit(stmt, lines, level + 1)
59
60     def _visit_Block(self, node: Block, lines: list[str], level: int)
61         -> None:
62         lines.append(f"{self._pad(level)}Block [{len(node.statements)}
63             statement(s)]")
64         for stmt in node.statements:
65             self._visit(stmt, lines, level + 1)
66
67     def _visit_Assign(self, node: Assign, lines: list[str], level: int)
68         -> None:
69         lines.append(f"{self._pad(level)}Assign {node.name!r} =")
70         self._visit(node.value, lines, level + 1)
71
72     def _visit_If(self, node: If, lines: list[str], level: int) -> None
73         :
74         lines.append(f"{self._pad(level)}If")
75         lines.append(f"{self._pad(level + 1)}condition:")
76         self._visit(node.condition, lines, level + 2)
77         lines.append(f"{self._pad(level + 1)}then:")
78         self._visit(node.then_block, lines, level + 2)
79         if node.else_block is not None:
80             lines.append(f"{self._pad(level + 1)}else:")
81             self._visit(node.else_block, lines, level + 2)
82
83     def _visit_While(self, node: While, lines: list[str], level: int)
84         -> None:
85         lines.append(f"{self._pad(level)}While")
86         lines.append(f"{self._pad(level + 1)}condition:")
87         self._visit(node.condition, lines, level + 2)
88         lines.append(f"{self._pad(level + 1)}body:")
89         self._visit(node.body, lines, level + 2)
90
91     def _visit_FunctionDef(self, node: FunctionDef, lines: list[str],

```

```

level: int) -> None:
85     params = ", ".join(node.params) if node.params else "(none)"
86     lines.append(f"{self._pad(level)}FunctionDef  {node.name!r}
           params=({params})")
87     self._visit(node.body, lines, level + 1)
88
89     def _visit_FunctionCall(self, node: FunctionCall, lines: list[str],
           level: int) -> None:
90         lines.append(f"{self._pad(level)}FunctionCall  {node.name!r}
           [{len(node.args)} arg(s)]")
91         for i, arg in enumerate(node.args):
92             lines.append(f"{self._pad(level + 1)}arg[{i}]:")
93             self._visit(arg, lines, level + 2)
94
95     def _visit_Return(self, node: Return, lines: list[str], level: int)
           -> None:
96         lines.append(f"{self._pad(level)}Return")
97         self._visit(node.expr, lines, level + 1)
98
99     def _visit_BinaryOp(self, node: BinaryOp, lines: list[str], level:
           int) -> None:
100         lines.append(f"{self._pad(level)}BinaryOp  '{node.op}'")
101         lines.append(f"{self._pad(level + 1)}left:")
102         self._visit(node.left, lines, level + 2)
103         lines.append(f"{self._pad(level + 1)}right:")
104         self._visit(node.right, lines, level + 2)
105
106     def _visit_Literal(self, node: Literal, lines: list[str], level:
           int) -> None:
107         type_name = type(node.value).__name__
108         lines.append(f"{self._pad(level)}Literal  {node.value!r}  ({
           type_name})")
109
110     def _visit_Identifier(self, node: Identifier, lines: list[str],
           level: int) -> None:
111         lines.append(f"{self._pad(level)}Identifier  {node.name!r}")
112
113     def _visit_unknown(self, node: ASTNode, lines: list[str], level:
           int) -> None:
114         lines.append(f"{self._pad(level)}??? Unknown node: {type(node)}.
           __name__")

```

Listing 8.8: components/ast\_printer.py — AST pretty-printer

```

1  from __future__ import annotations
2
3  from dataclasses import dataclass, field
4  from enum import Enum
5
6
7  class LanguageType(Enum):
8      INTEGER = "Integer"
9      FLOAT = "Float"

```

```

10     BOOLEAN = "Boolean"
11     STRING = "String"
12     VOID = "Void"
13
14
15 @dataclass
16 class Symbol:
17     name: str
18     symbol_type: LanguageType
19
20
21 @dataclass
22 class VariableSymbol(Symbol):
23     initialized: bool = False
24
25
26 @dataclass
27 class FunctionSymbol(Symbol):
28     parameters: list[tuple[str, LanguageType]] = field(default_factory=
        list)
29
30
31 class SymbolTableError(Exception):
32     pass
33
34
35 class SymbolTable:
36     def __init__(self, parent: SymbolTable | None = None) -> None:
37         self.parent = parent
38         self._symbols: dict[str, Symbol] = {}
39
40     def define_variable(self, name: str, symbol_type: LanguageType,
        initialized: bool = False) -> VariableSymbol:
41         if name in self._symbols:
42             raise SymbolTableError(f"Symbol already defined in this
                scope: {name}")
43         symbol = VariableSymbol(name=name, symbol_type=symbol_type,
            initialized=initialized)
44         self._symbols[name] = symbol
45         return symbol
46
47     def define_function(
48         self,
49         name: str,
50         return_type: LanguageType,
51         parameters: list[tuple[str, LanguageType]],
52     ) -> FunctionSymbol:
53         if name in self._symbols:
54             raise SymbolTableError(f"Symbol already defined in this
                scope: {name}")
55         symbol = FunctionSymbol(name=name, symbol_type=return_type,
            parameters=parameters)
56         self._symbols[name] = symbol

```

```

57         return symbol
58
59     def lookup(self, name: str) -> Symbol:
60         symbol = self._symbols.get(name)
61         if symbol is not None:
62             return symbol
63         if self.parent is not None:
64             return self.parent.lookup(name)
65         raise SymbolTableError(f"Undefined symbol: {name}")
66
67     def exists_in_current_scope(self, name: str) -> bool:
68         return name in self._symbols
69
70     def child_scope(self) -> SymbolTable:
71         return SymbolTable(parent=self)
72
73     def set_initialized(self, name: str) -> None:
74         symbol = self.lookup(name)
75         if not isinstance(symbol, VariableSymbol):
76             raise SymbolTableError(f"Cannot initialize non-variable
77                                     symbol: {name}")
78         symbol.initialized = True
79
80     def snapshot(self) -> dict[str, Symbol]:
81         return dict(self._symbols)
82
83     def format_table(self) -> str:
84         lines = [
85             f"{'Name':<12} {'Kind':<10} {'Type':<10} {'Initialized'
86               ':<12} Parameters",
87             "-" * 72,
88         ]
89
90         for name, symbol in self._symbols.items():
91             if isinstance(symbol, FunctionSymbol):
92                 params = ", ".join(f"{param_name}: {param_type.value}"
93                                     for param_name, param_type in symbol.parameters)
94                 lines.append(
95                     f"{name:<12} {'function':<10} {symbol.symbol_type.
96                       value:<10} {'-':<12} {params or '-'}"
97                 )
98             elif isinstance(symbol, VariableSymbol):
99                 initialized = "yes" if symbol.initialized else "no"
100                 lines.append(
101                     f"{name:<12} {'variable':<10} {symbol.symbol_type.
102                       value:<10} {initialized:<12} -"
103                 )
104             else:
105                 lines.append(f"{name:<12} {'symbol':<10} {symbol.
106                               symbol_type.value:<10} {'-':<12} -")
107
108         return "\n".join(lines)

```

Listing 8.9: components/symbol\_table.py — scoped symbol table

```
1 from __future__ import annotations
2
3 from dataclasses import dataclass
4
5 from components.ast_nodes import (
6     ASTNode,
7     Assign,
8     BinaryOp,
9     Block,
10    FunctionCall,
11    FunctionDef,
12    Identifier,
13    If,
14    Literal,
15    Program,
16    Return,
17    While,
18 )
19 from components.symbol_table import FunctionSymbol, LanguageType,
20     SymbolTable, SymbolTableError, VariableSymbol
21
22 class TypeCheckError(Exception):
23     pass
24
25
26 @dataclass
27 class _FunctionState:
28     node: FunctionDef
29     body_checked: bool = False
30
31
32 @dataclass
33 class _FunctionContext:
34     name: str
35     return_type: LanguageType | None = None
36
37
38 class TypeChecker:
39     _BUILTIN_PRINT = "print"
40
41     def __init__(self) -> None:
42         self.global_scope = SymbolTable()
43         self._functions: dict[str, _FunctionState] = {}
44         self._active_calls: list[str] = []
45
46     def check(self, program: Program) -> SymbolTable:
47         for statement in program.statements:
48             if isinstance(statement, FunctionDef):
49                 self._register_function(statement)
```

```

50
51     for statement in program.statements:
52         if isinstance(statement, FunctionDef):
53             continue
54         self._check_statement(statement, self.global_scope, None)
55
56     self._check_uncalled_functions()
57     return self.global_scope
58
59 def _register_function(self, node: FunctionDef) -> None:
60     if node.name in self._functions:
61         raise TypeCheckError(self._error_at(node, f"Function '{node
62             .name}' is already defined"))
63
64     placeholder_parameters = [(param_name, LanguageType.VOID) for
65         param_name in node.params]
66     try:
67         self.global_scope.define_function(node.name, LanguageType.
68             VOID, placeholder_parameters)
69     except SymbolTableError as error:
70         raise TypeCheckError(self._error_at(node, str(error))) from
71             error
72     self._functions[node.name] = _FunctionState(node=node)
73
74 def _check_statement(
75     self,
76     node: ASTNode,
77     scope: SymbolTable,
78     function_context: _FunctionContext | None,
79 ) -> None:
80     if isinstance(node, Assign):
81         value_type = self._infer_expr_type(node.value, scope)
82         self._assign_variable_type(scope, node, value_type)
83         return
84
85     if isinstance(node, Return):
86         if function_context is None:
87             raise TypeCheckError(self._error_at(node, "'return' is
88                 only valid inside a function"))
89         expr_type = self._infer_expr_type(node.expr, scope)
90         if function_context.return_type is None:
91             function_context.return_type = expr_type
92         elif function_context.return_type != expr_type:
93             raise TypeCheckError(
94                 self._error_at(
95                     node,
96                     f"Function '{function_context.name}' returns
97                         both {function_context.return_type.value}
98                         and {expr_type.value}",
99                 )
100             )
101         return

```

```

96     if isinstance(node, If):
97         condition_type = self._infer_expr_type(node.condition,
98             scope)
99         if condition_type != LanguageType.BOOLEAN:
100             raise TypeCheckError(self._error_at(node.condition, "If
101                 condition must have type Boolean"))
102         self._check_block(node.then_block, scope.child_scope(),
103             function_context)
104         if node.else_block is not None:
105             self._check_block(node.else_block, scope.child_scope(),
106                 function_context)
107         return
108
109     if isinstance(node, While):
110         condition_type = self._infer_expr_type(node.condition,
111             scope)
112         if condition_type != LanguageType.BOOLEAN:
113             raise TypeCheckError(self._error_at(node.condition, "
114                 While condition must have type Boolean"))
115         self._check_block(node.body, scope.child_scope(),
116             function_context)
117         return
118
119     if isinstance(node, Block):
120         self._check_block(node, scope.child_scope(),
121             function_context)
122         return
123
124     if isinstance(node, FunctionCall):
125         self._infer_function_call_type(node, scope)
126         return
127
128     raise TypeCheckError(self._error_at(node, f"Unsupported
129         statement node: {type(node).__name__}"))
130
131 def _check_block(
132     self,
133     block: Block,
134     scope: SymbolTable,
135     function_context: _FunctionContext | None,
136 ) -> None:
137     for statement in block.statements:
138         self._check_statement(statement, scope, function_context)
139
140 def _infer_expr_type(self, node: ASTNode, scope: SymbolTable) ->
141     LanguageType:
142     if isinstance(node, Literal):
143         return self._literal_type(node)
144
145     if isinstance(node, Identifier):
146         try:
147             symbol = scope.lookup(node.name)
148         except SymbolTableError as error:

```

```

139         raise TypeCheckError(self._error_at(node, str(error)))
140         from error
141     if not isinstance(symbol, VariableSymbol):
142         raise TypeCheckError(self._error_at(node, f"'{node.name
143         }' is not a variable"))
144     return symbol.symbol_type
145
146 if isinstance(node, FunctionCall):
147     return self._infer_function_call_type(node, scope)
148
149 if isinstance(node, BinaryOp):
150     left_type = self._infer_expr_type(node.left, scope)
151     right_type = self._infer_expr_type(node.right, scope)
152     return self._infer_binary_type(node, left_type, right_type)
153
154 raise TypeCheckError(self._error_at(node, f"Unsupported
155 expression node: {type(node).__name__}"))
156
157 def _infer_function_call_type(self, node: FunctionCall, scope:
158 SymbolTable) -> LanguageType:
159     if node.name == self._BUILTIN_PRINT:
160         return self._infer_builtin_print_type(node, scope)
161
162     function_state = self._functions.get(node.name)
163     if function_state is None:
164         raise TypeCheckError(self._error_at(node, f"Undefined
165         function: {node.name}"))
166     if node.name in self._active_calls:
167         raise TypeCheckError(self._error_at(node, f"Recursive
168         function calls are not supported: {node.name}"))
169
170     argument_types = [self._infer_expr_type(argument, scope) for
171     argument in node.args]
172     function_symbol = self._lookup_function_symbol(node)
173
174     if len(argument_types) != len(function_symbol.parameters):
175         raise TypeCheckError(
176             self._error_at(
177                 node,
178                 f"Function '{node.name}' expects {len(
179                     function_symbol.parameters)} argument(s), got {
180                     len(argument_types)}",
181             )
182         )
183
184     declared_parameter_types = [param_type for _, param_type in
185     function_symbol.parameters]
186     if all(param_type == LanguageType.VOID for param_type in
187     declared_parameter_types):
188         function_symbol.parameters = list(zip(function_state.node.
189         params, argument_types, strict=False))
190     else:
191         for index, ((_, expected_type), actual_type) in enumerate(

```

```

        zip(function_symbol.parameters, argument_types, strict=
        False)):
180         if expected_type != actual_type:
181             raise TypeCheckError(
182                 self._error_at(
183                     node.args[index],
184                     f"Argument {index + 1} of '{node.name}'
                        must be {expected_type.value}, got {
                        actual_type.value}",
185                     )
186             )
187
188     if not function_state.body_checked:
189         self._check_function_body(function_state, function_symbol)
190
191     return function_symbol.symbol_type
192
193 def _infer_builtin_print_type(self, node: FunctionCall, scope:
    SymbolTable) -> LanguageType:
194     if len(node.args) != 1:
195         raise TypeCheckError(
196             self._error_at(node, f"Built-in function 'print'
                        expects 1 argument, got {len(node.args)}")
197         )
198     self._infer_expr_type(node.args[0], scope)
199     return LanguageType.VOID
200
201 def _check_function_body(self, function_state: _FunctionState,
    function_symbol: FunctionSymbol) -> None:
202     function_context = _FunctionContext(name=function_state.node.
        name)
203     function_scope = self.global_scope.child_scope()
204     for parameter_name, parameter_type in function_symbol.
        parameters:
205         function_scope.define_variable(parameter_name,
            parameter_type, initialized=True)
206
207     self._active_calls.append(function_state.node.name)
208     try:
209         self._check_block(function_state.node.body, function_scope,
            function_context)
210     finally:
211         self._active_calls.pop()
212
213     function_symbol.symbol_type = function_context.return_type or
        LanguageType.VOID
214     function_state.body_checked = True
215
216 def _check_uncalled_functions(self) -> None:
217     """After all call-site checks, type-check any function whose
        body was never visited."""
218     for name, state in self._functions.items():
219         if not state.body_checked:

```

```

220         try:
221             function_symbol = self.global_scope.lookup(name)
222         except SymbolTableError:
223             continue
224         if not isinstance(function_symbol, FunctionSymbol):
225             continue
226         inferred_params = self._infer_param_types_from_body(
227             state.node)
228         function_symbol.parameters = inferred_params
229         self._check_function_body(state, function_symbol)
230
231     def _infer_param_types_from_body(self, node: FunctionDef) -> list[
232         tuple[str, LanguageType]]:
233         """Infer parameter types by scanning the body for expression
234         contexts.
235
236         Any parameter used as an operand of a float operator (+, -, *,
237         /.) is
238         inferred as Float; a parameter used with an integer operator (+
239         - * /
240         == !=) is inferred as Integer. Parameters with no type hint
241         default to
242         Integer.
243         """
244         param_names = set(node.params)
245         inferred: dict[str, LanguageType] = {}
246
247     def scan_expr(n: ASTNode) -> None:
248         if isinstance(n, BinaryOp):
249             if n.op in {"+", "-", "*", "/", "==", "!="}:
250                 for operand in (n.left, n.right):
251                     if isinstance(operand, Identifier) and operand.
252                         name in param_names:
253                         inferred.setdefault(operand.name,
254                             LanguageType.INTEGER)
255             elif n.op in {"+.", "-.", "*.", "/."}:
256                 for operand in (n.left, n.right):
257                     if isinstance(operand, Identifier) and operand.
258                         name in param_names:
259                         inferred.setdefault(operand.name,
260                             LanguageType.FLOAT)
261             scan_expr(n.left)
262             scan_expr(n.right)
263         elif isinstance(n, FunctionCall):
264             for arg in n.args:
265                 scan_expr(arg)
266
267     def scan_stmt(n: ASTNode) -> None:
268         if isinstance(n, Assign):
269             scan_expr(n.value)
270         elif isinstance(n, Return):
271             scan_expr(n.expr)
272         elif isinstance(n, If):

```

```

263         scan_expr(n.condition)
264         scan_block(n.then_block)
265         if n.else_block:
266             scan_block(n.else_block)
267     elif isinstance(n, While):
268         scan_expr(n.condition)
269         scan_block(n.body)
270     elif isinstance(n, Block):
271         scan_block(n)
272     elif isinstance(n, FunctionCall):
273         for arg in n.args:
274             scan_expr(arg)
275
276 def scan_block(block: Block) -> None:
277     for stmt in block.statements:
278         scan_stmt(stmt)
279
280 scan_block(node.body)
281 return [(param, inferred.get(param, LanguageType.INTEGER)) for
282         param in node.params]
283
284 def _assign_variable_type(self, scope: SymbolTable, node: Assign,
285 value_type: LanguageType) -> None:
286     if value_type == LanguageType.VOID:
287         raise TypeCheckError(self._error_at(node, f"Cannot assign
288 the result of a void function to '{node.name}'))
289     try:
290         symbol = scope.lookup(node.name)
291     except SymbolTableError:
292         scope.define_variable(node.name, value_type, initialized=
293 True)
294     return
295
296 if not isinstance(symbol, VariableSymbol):
297     raise TypeCheckError(self._error_at(node, f"'{node.name}'
298 is not a variable"))
299 if symbol.symbol_type != value_type:
300     raise TypeCheckError(
301 self._error_at(
302 node,
303 f"Cannot assign {value_type.value} to variable '{
304 node.name}' of type {symbol.symbol_type.value}",
305 )
306 )
307 symbol.initialized = True
308
309 def _infer_binary_type(
310 self,
311 node: BinaryOp,
312 left_type: LanguageType,
313 right_type: LanguageType,
314 ) -> LanguageType:
315     # Integer operators: both operands must be Integer

```

```

310     if node.op in {"+", "-", "*"}:
311         if left_type != LanguageType.INTEGER or right_type !=
           LanguageType.INTEGER:
312             raise TypeCheckError(
313                 self._error_at(node, f"Operator '{node.op}'
                     requires two Integer operands")
           )
314         return LanguageType.INTEGER
315
316
317     # Integer division: both operands must be Integer, result is
           Integer
318     if node.op == "/":
319         if left_type != LanguageType.INTEGER or right_type !=
           LanguageType.INTEGER:
320             raise TypeCheckError(
321                 self._error_at(node, f"Operator '{node.op}'
                     requires two Integer operands")
           )
322         return LanguageType.INTEGER
323
324
325     # Float operators: both operands must be Float
326     if node.op in {"+.", "-.", "*.", "/."}:
327         if left_type != LanguageType.FLOAT or right_type !=
           LanguageType.FLOAT:
328             raise TypeCheckError(
329                 self._error_at(node, f"Operator '{node.op}'
                     requires two Float operands")
           )
330         return LanguageType.FLOAT
331
332
333     if node.op in {"==", "!="}:
334         if not self._is_numeric(left_type) or not self._is_numeric(
           right_type):
335             raise TypeCheckError(
336                 self._error_at(
337                     node,
338                     f"Operator '{node.op}' requires two arithmetic
                         operands",
339                 )
340             )
341         if left_type != right_type:
342             raise TypeCheckError(
343                 self._error_at(
344                     node,
345                     f"Operator '{node.op}' requires operands of the
                         same type, got {left_type.value} and {
                             right_type.value}",
346                 )
347             )
348         return LanguageType.BOOLEAN
349
350     raise TypeCheckError(self._error_at(node, f"Unsupported
           operator: {node.op}"))

```

```

351
352 def _lookup_function_symbol(self, node: FunctionCall) ->
    FunctionSymbol:
353     try:
354         symbol = self.global_scope.lookup(node.name)
355     except SymbolTableError as error:
356         raise TypeCheckError(self._error_at(node, str(error))) from
            error
357     if not isinstance(symbol, FunctionSymbol):
358         raise TypeCheckError(self._error_at(node, f"'{node.name}',
            is not a function"))
359     return symbol
360
361 @staticmethod
362 def _is_numeric(language_type: LanguageType) -> bool:
363     return language_type in {LanguageType.INTEGER, LanguageType.
        FLOAT}
364
365 @staticmethod
366 def _literal_type(node: Literal) -> LanguageType:
367     if isinstance(node.value, bool):
368         return LanguageType.BOOLEAN
369     if isinstance(node.value, int):
370         return LanguageType.INTEGER
371     if isinstance(node.value, float):
372         return LanguageType.FLOAT
373     if isinstance(node.value, str):
374         return LanguageType.STRING
375     raise TypeCheckError(f"Unsupported literal value: {node.value!r}
        ")
376
377 @staticmethod
378 def _error_at(node: ASTNode, message: str) -> str:
379     return f"[line {node.line}, col {node.column}] TypeError: {
        message}"

```

Listing 8.10: components/type\_checker.py — static type checker with inference

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass
4 from typing import Callable
5
6 from components.ast_nodes import (
7     ASTNode,
8     Assign,
9     BinaryOp,
10    Block,
11    FunctionCall,
12    FunctionDef,
13    Identifier,
14    If,
15    Literal,

```

```

16     Program,
17     Return,
18     While,
19 )
20 from components.symbol_table import LanguageType
21
22
23 class InterpreterError(Exception):
24     pass
25
26
27 class _ReturnSignal(Exception):
28     def __init__(self, value: object) -> None:
29         self.value = value
30
31
32 @dataclass
33 class _FunctionRuntime:
34     node: FunctionDef
35
36
37 class _Environment:
38     def __init__(self, parent: _Environment | None = None) -> None:
39         self.parent = parent
40         self.values: dict[str, object] = {}
41
42     def define(self, name: str, value: object) -> None:
43         self.values[name] = value
44
45     def get(self, name: str) -> object:
46         if name in self.values:
47             return self.values[name]
48         if self.parent is not None:
49             return self.parent.get(name)
50         raise InterpreterError(f"Undefined variable: {name}")
51
52     def assign(self, name: str, value: object) -> None:
53         if name in self.values:
54             self.values[name] = value
55             return
56         if self.parent is not None and self.parent.contains(name):
57             self.parent.assign(name, value)
58             return
59         self.values[name] = value
60
61     def contains(self, name: str) -> bool:
62         if name in self.values:
63             return True
64         if self.parent is not None:
65             return self.parent.contains(name)
66         return False
67
68

```

```

69 class Interpreter:
70     _BUILTIN_PRINT = "print"
71
72     def __init__(self, output_callback: Callable[[str], None] | None =
73         None) -> None:
74         self._functions: dict[str, _FunctionRuntime] = {}
75         self._output_callback = output_callback or print
76
77     def execute(self, program: Program) -> None:
78         environment = _Environment()
79         for statement in program.statements:
80             if isinstance(statement, FunctionDef):
81                 self._functions[statement.name] = _FunctionRuntime(node
82                     =statement)
83
84         for statement in program.statements:
85             if isinstance(statement, FunctionDef):
86                 continue
87             self._execute_statement(statement, environment)
88
89     def _execute_statement(self, node: ASTNode, environment:
90         _Environment) -> None:
91         if isinstance(node, Assign):
92             value = self._evaluate(node.value, environment)
93             environment.assign(node.name, value)
94             return
95
96         if isinstance(node, Return):
97             raise _ReturnSignal(self._evaluate(node.expr, environment))
98
99         if isinstance(node, If):
100             condition = self._evaluate(node.condition, environment)
101             if not isinstance(condition, bool):
102                 raise InterpreterError(self._error_at(node.condition, "
103                     If condition must evaluate to Boolean"))
104             branch = node.then_block if condition else node.else_block
105             if branch is not None:
106                 self._execute_block(branch, _Environment(parent=
107                     environment))
108             return
109
110         if isinstance(node, While):
111             while True:
112                 condition = self._evaluate(node.condition, environment)
113                 if not isinstance(condition, bool):
114                     raise InterpreterError(self._error_at(node.
115                         condition, "While condition must evaluate to
116                         Boolean"))
117                 if not condition:
118                     break
119                 self._execute_block(node.body, _Environment(parent=
120                     environment))
121             return

```

```

114
115     if isinstance(node, Block):
116         self._execute_block(node, _Environment(parent=environment))
117         return
118
119     if isinstance(node, FunctionCall):
120         self._evaluate(node, environment)
121         return
122
123     raise InterpreterError(self._error_at(node, f"Unsupported
        statement node: {type(node).__name__}"))
124
125 def _execute_block(self, block: Block, environment: _Environment)
-> None:
126     for statement in block.statements:
127         self._execute_statement(statement, environment)
128
129 def _evaluate(self, node: ASTNode, environment: _Environment) ->
object:
130     if isinstance(node, Literal):
131         return node.value
132
133     if isinstance(node, Identifier):
134         return environment.get(node.name)
135
136     if isinstance(node, BinaryOp):
137         left_value = self._evaluate(node.left, environment)
138         right_value = self._evaluate(node.right, environment)
139         return self._evaluate_binary(node, left_value, right_value)
140
141     if isinstance(node, FunctionCall):
142         return self._call_function(node, environment)
143
144     raise InterpreterError(self._error_at(node, f"Unsupported
        expression node: {type(node).__name__}"))
145
146 def _call_function(self, node: FunctionCall, environment:
_Environment) -> object:
147     if node.name == self._BUILTIN_PRINT:
148         return self._call_builtin_print(node, environment)
149
150     function_runtime = self._functions.get(node.name)
151     if function_runtime is None:
152         raise InterpreterError(self._error_at(node, f"Undefined
            function: {node.name}"))
153
154     if len(node.args) != len(function_runtime.node.params):
155         raise InterpreterError(
156             self._error_at(
157                 node,
158                 f"Function '{node.name}' expects {len(
                    function_runtime.node.params)} argument(s), got
                    {len(node.args)}",

```

```

159         )
160     )
161
162     call_environment = _Environment(parent=environment)
163     argument_values = [self._evaluate(argument, environment) for
164         argument in node.args]
165     for parameter_name, argument_value in zip(function_runtime.node
166         .params, argument_values, strict=False):
167         call_environment.define(parameter_name, argument_value)
168
169     try:
170         self._execute_block(function_runtime.node.body,
171             call_environment)
172     except _ReturnSignal as signal:
173         return signal.value
174     return None
175
176 def _call_builtin_print(self, node: FunctionCall, environment:
177     _Environment) -> None:
178     if len(node.args) != 1:
179         raise InterpreterError(
180             self._error_at(node, f"Built-in function 'print'
181                 expects 1 argument, got {len(node.args)}")
182         )
183     value = self._evaluate(node.args[0], environment)
184     self._output_callback(f"{self._format_value(value)} : {self.
185         _runtime_type(value).value}")
186     return None
187
188 def _evaluate_binary(self, node: BinaryOp, left_value: object,
189     right_value: object) -> object:
190     if node.op == "+":
191         return left_value + right_value
192     if node.op == "-":
193         return left_value - right_value
194     if node.op == "*":
195         return left_value * right_value
196     if node.op == "/":
197         if int(right_value) == 0:
198             raise InterpreterError(self._error_at(node, "Division
199                 by zero"))
200         return int(left_value) // int(right_value)
201     if node.op == "+.":
202         return float(left_value) + float(right_value)
203     if node.op == "-.":
204         return float(left_value) - float(right_value)
205     if node.op == ".*":
206         return float(left_value) * float(right_value)
207     if node.op == "/.":
208         if float(right_value) == 0.0:
209             raise InterpreterError(self._error_at(node, "Division
210                 by zero"))
211         return float(left_value) / float(right_value)

```

```

203     if node.op == "==":
204         return left_value == right_value
205     if node.op == "!=":
206         return left_value != right_value
207     raise InterpreterError(self._error_at(node, f"Unsupported
        operator: {node.op}"))
208
209     @staticmethod
210     def _format_value(value: object) -> str:
211         if isinstance(value, str):
212             return f'"{value}"'
213         if isinstance(value, bool):
214             return "true" if value else "false"
215         return str(value)
216
217     @staticmethod
218     def _runtime_type(value: object) -> LanguageType:
219         if isinstance(value, bool):
220             return LanguageType.BOOLEAN
221         if isinstance(value, int):
222             return LanguageType.INTEGER
223         if isinstance(value, float):
224             return LanguageType.FLOAT
225         if isinstance(value, str):
226             return LanguageType.STRING
227         return LanguageType.VOID
228
229     @staticmethod
230     def _error_at(node: ASTNode, message: str) -> str:
231         return f"[line {node.line}, col {node.column}] RuntimeError: {
            message}"

```

Listing 8.11: components/interpreter.py — tree-walking interpreter

```

1  from __future__ import annotations
2
3  from dataclasses import dataclass
4
5  from components.ast_printer import ASTPrinter
6  from components.interpreter import Interpreter
7  from components.lexica import Lexer
8  from components.parse_tree_printer import ParseTreePrinter
9  from components.parser import Parser
10 from components.symbol_table import SymbolTable
11 from components.tokens import Token
12 from components.type_checker import TypeChecker
13
14
15 @dataclass
16 class PipelineResult:
17     tokens: list[Token]
18     parse_tree_text: str
19     ast_text: str

```

```

20     checked_scope: SymbolTable
21     outputs: list[str]
22
23
24 def run_pipeline(source_code: str) -> PipelineResult:
25     tokens = Lexer(source_code).tokenize()
26     tree = Parser(tokens).parse()
27     parse_tree_text = ParseTreePrinter().print(tree)
28     ast_text = ASTPrinter().print(tree)
29     checked_scope = TypeChecker().check(tree)
30
31     outputs: list[str] = []
32     Interpreter(output_callback=outputs.append).execute(tree)
33
34     return PipelineResult(
35         tokens=tokens,
36         parse_tree_text=parse_tree_text,
37         ast_text=ast_text,
38         checked_scope=checked_scope,
39         outputs=outputs,
40     )
41
42
43 def format_tokens(tokens: list[Token]) -> str:
44     lines = []
45     for token in tokens:
46         lines.append(
47             f"    {token.token_type.name:<14} {token.lexeme!r:<12} line={
48                 token.line} col={token.column}"
49         )
50     return "\n".join(lines)
51
52 def format_stage_output(result: PipelineResult) -> str:
53     sections = [
54         "=" * 60,
55         "    STAGE 1 -- TOKENS",
56         "=" * 60,
57         format_tokens(result.tokens),
58         "",
59         "=" * 60,
60         "    STAGE 2 -- PARSE TREE",
61         "=" * 60,
62         result.parse_tree_text,
63         "",
64         "=" * 60,
65         "    STAGE 3 -- AST",
66         "=" * 60,
67         result.ast_text,
68         "",
69         "=" * 60,
70         "    STAGE 4 -- TYPE CHECK",
71         "=" * 60,

```

```

72     result.checked_scope.format_table(),
73     "",
74     "=" * 60,
75     "    STAGE 5 -- EXECUTION",
76     "=" * 60,
77     "\n".join(result.outputs) if result.outputs else "(no output)",
78 ]
79 return "\n".join(sections)

```

Listing 8.12: components/pipeline.py — shared pipeline (CLI, UI, and tests)

### A.3 Test Suite

```

1 from __future__ import annotations
2
3 import unittest
4
5 from components.interpreter import InterpreterError
6 from components.lexica import Lexer, LexerError
7 from components.parser import Parser, ParseError
8 from components.pipeline import run_pipeline
9 from components.symbol_table import LanguageType, SymbolTable,
   SymbolTableError
10 from components.tokens import TokenType
11 from components.type_checker import TypeCheckError
12
13
14 #
   -----
15 # Lexer
16 #
   -----
17
18 class LexerTests(unittest.TestCase):
19     """Tests for components/lexica.py -- lexical analysis and
   tokenisation."""
20
21     def test_integer_literal_token(self) -> None:
22         tokens = Lexer("42").tokenize()
23         self.assertEqual(tokens[0].token_type, TokenType.INT_LITERAL)
24         self.assertEqual(tokens[0].lexeme, "42")
25
26     def test_float_literal_token(self) -> None:
27         tokens = Lexer("3.14").tokenize()
28         self.assertEqual(tokens[0].token_type, TokenType.FLOAT_LITERAL)
29         self.assertEqual(tokens[0].lexeme, "3.14")
30
31     def test_bool_literals_produce_bool_literal_tokens(self) -> None:
32         tokens = Lexer("true false").tokenize()
33         self.assertEqual(tokens[0].token_type, TokenType.BOOL_LITERAL)

```

```

34     self.assertEqual(tokens[0].lexeme, "true")
35     self.assertEqual(tokens[1].token_type, TokenType.BOOL_LITERAL)
36     self.assertEqual(tokens[1].lexeme, "false")
37
38     def test_keywords_produce_correct_token_types(self) -> None:
39         tokens = Lexer("if else while def return").tokenize()
40         expected = [
41             TokenType.IF, TokenType.ELSE, TokenType.WHILE, TokenType.
42             DEF, TokenType.RETURN,
43         ]
44         actual = [t.token_type for t in tokens if t.token_type !=
45             TokenType.EOF]
46         self.assertEqual(actual, expected)
47
48     def test_identifier_not_confused_with_keyword(self) -> None:
49         tokens = Lexer("iffy whileloop print").tokenize()
50         self.assertEqual(tokens[0].token_type, TokenType.IDENTIFIER)
51         self.assertEqual(tokens[1].token_type, TokenType.IDENTIFIER)
52         self.assertEqual(tokens[2].token_type, TokenType.IDENTIFIER)
53
54     def test_two_character_operators(self) -> None:
55         tokens = Lexer("== !=").tokenize()
56         self.assertEqual(tokens[0].token_type, TokenType.EQUAL_EQUAL)
57         self.assertEqual(tokens[1].token_type, TokenType.BANG_EQUAL)
58
59     def test_float_dot_operators_tokenised(self) -> None:
60         tokens = Lexer("+. -. *. /.").tokenize()
61         expected = [TokenType.PLUS_DOT, TokenType.MINUS_DOT, TokenType.
62             STAR_DOT, TokenType.SLASH_DOT]
63         actual = [t.token_type for t in tokens if t.token_type !=
64             TokenType.EOF]
65         self.assertEqual(actual, expected)
66
67     def test_float_literal_not_confused_with_dot_operator(self) -> None
68     :
69         # "3.14" must produce FLOAT_LITERAL, not INT_LITERAL followed
70         by something
71         tokens = Lexer("3.14").tokenize()
72         self.assertEqual(tokens[0].token_type, TokenType.FLOAT_LITERAL)
73         self.assertEqual(tokens[0].lexeme, "3.14")
74
75     def test_source_position_tracked_across_lines(self) -> None:
76         tokens = Lexer("x\ny").tokenize()
77         self.assertEqual(tokens[0].line, 1)
78         self.assertEqual(tokens[1].line, 2)
79
80     def test_unexpected_character_raises_lexer_error(self) -> None:
81         with self.assertRaises(LexerError):
82             Lexer("@").tokenize()
83
84     def test_lone_exclamation_raises_lexer_error(self) -> None:
85         with self.assertRaises(LexerError):
86             Lexer("!5").tokenize()

```

```

81
82     def test_terminated_string_raises_lexer_error(self) -> None:
83         with self.assertRaises(LexerError):
84             Lexer('"hello"').tokenize()
85
86
87     #
88     -----
89     # Symbol table
90     #
91     -----
92
93
94     class SymbolTableTests(unittest.TestCase):
95         """Tests for components/symbol_table.py -- scoped symbol table and
96         semantic types."""
97
98         def test_define_variable_and_lookup(self) -> None:
99             table = SymbolTable()
100             table.define_variable("x", LanguageType.INTEGER, initialized=
101                 True)
102             symbol = table.lookup("x")
103             self.assertEqual(symbol.symbol_type, LanguageType.INTEGER)
104
105         def test_duplicate_variable_definition_raises_error(self) -> None:
106             table = SymbolTable()
107             table.define_variable("x", LanguageType.INTEGER)
108             with self.assertRaises(SymbolTableError):
109                 table.define_variable("x", LanguageType.FLOAT)
110
111         def test_duplicate_function_definition_raises_error(self) -> None:
112             table = SymbolTable()
113             table.define_function("f", LanguageType.INTEGER, [])
114             with self.assertRaises(SymbolTableError):
115                 table.define_function("f", LanguageType.FLOAT, [])
116
117         def test_lookup_in_parent_scope(self) -> None:
118             parent = SymbolTable()
119             parent.define_variable("x", LanguageType.INTEGER, initialized=
120                 True)
121             child = parent.child_scope()
122             symbol = child.lookup("x")
123             self.assertEqual(symbol.symbol_type, LanguageType.INTEGER)
124
125         def test_undefined_symbol_raises_error(self) -> None:
126             table = SymbolTable()
127             with self.assertRaises(SymbolTableError):
128                 table.lookup("z")
129
130         def test_format_table_contains_variable_row(self) -> None:
131             table = SymbolTable()

```

```

126         table.define_variable("counter", LanguageType.INTEGER,
127                               initialized=True)
128         output = table.format_table()
129         self.assertIn("counter", output)
130         self.assertIn("variable", output)
131         self.assertIn("Integer", output)
132
133     def test_format_table_contains_function_row(self) -> None:
134         table = SymbolTable()
135         table.define_function("add", LanguageType.INTEGER, [("a",
136                                                             LanguageType.INTEGER)])
137         output = table.format_table()
138         self.assertIn("add", output)
139         self.assertIn("function", output)
140
141     #
142     -----
143
144     # Parser
145     #
146     -----
147
148 class ParserTests(unittest.TestCase):
149     """Tests for components/parser.py -- recursive-descent parsing."""
150
151     def _parse(self, source: str):
152         return Parser(Lexer(source).tokenize()).parse()
153
154     def test_missing_semicolon_raises_parse_error(self) -> None:
155         with self.assertRaises(ParseError):
156             self._parse("x = 5")
157
158     def test_unclosed_brace_raises_parse_error(self) -> None:
159         with self.assertRaises(ParseError):
160             self._parse("if (true) { x = 1;")
161
162     def test_operator_precedence_mul_before_add(self) -> None:
163         # 2 + 3 * 4 must be 14, not 20
164         result = run_pipeline("x = 2 + 3 * 4; print(x);")
165         self.assertEqual(result.outputs, ["14 : Integer"])
166
167     def test_operator_left_associativity(self) -> None:
168         # 10 - 3 - 2 must be 5 (left-assoc), not 9
169         result = run_pipeline("x = 10 - 3 - 2; print(x);")
170         self.assertEqual(result.outputs, ["5 : Integer"])
171
172     def test_parentheses_override_precedence(self) -> None:
173         # (2 + 3) * 4 must be 20
174         result = run_pipeline("x = (2 + 3) * 4; print(x);")
175         self.assertEqual(result.outputs, ["20 : Integer"])

```

```

173 def test_float_operator_precedence_mul_before_add(self) -> None:
174     # 1.0 +. 2.0 *. 3.0 must be 7.0 (*. binds tighter than +.)
175     result = run_pipeline("x = 1.0 +. 2.0 *. 3.0; print(x);")
176     self.assertEqual(result.outputs, ["7.0 : Float"])
177
178 def test_pipeline_exposes_parse_tree_and_ast(self) -> None:
179     result = run_pipeline("x = 5; print(x);")
180     self.assertIn("program", result.parse_tree_text)
181     self.assertIn("assignment", result.parse_tree_text)
182     self.assertIn("additive", result.parse_tree_text)
183     self.assertIn("term", result.parse_tree_text)
184     self.assertIn("factor", result.parse_tree_text)
185     self.assertIn("function_call", result.parse_tree_text)
186     self.assertIn("Program", result.ast_text)
187     self.assertIn("Assign", result.ast_text)
188     self.assertIn("FunctionCall 'print'", result.ast_text)
189
190
191 #
192 # -----
193 # Type checker
194 # -----
195
196 class TypeCheckerTests(unittest.TestCase):
197     """Tests for components/type_checker.py -- static type inference and
198     checking."""
199
200     def test_integer_arithmetic_stays_integer(self) -> None:
201         result = run_pipeline("x = 3 + 4; print(x);")
202         self.assertEqual(result.outputs, ["7 : Integer"])
203
204     def test_division_always_produces_integer(self) -> None:
205         # integer / integer -> Integer (no implicit float conversion)
206         result = run_pipeline("x = 10 / 2; print(x);")
207         self.assertEqual(result.outputs, ["5 : Integer"])
208
209     def test_float_dot_operators_produce_float(self) -> None:
210         result = run_pipeline("x = 1.0 +. 2.5; print(x);")
211         self.assertEqual(result.outputs, ["3.5 : Float"])
212
213     def test_float_subtraction_produces_float(self) -> None:
214         result = run_pipeline("x = 5.0 -. 1.5; print(x);")
215         self.assertEqual(result.outputs, ["3.5 : Float"])
216
217     def test_float_multiplication_produces_float(self) -> None:
218         result = run_pipeline("x = 2.0 *. 3.0; print(x);")
219         self.assertEqual(result.outputs, ["6.0 : Float"])
220
221     def test_float_division_produces_float(self) -> None:
222         result = run_pipeline("x = 9.0 /. 4.0; print(x);")

```

```

221         self.assertEqual(result.outputs, ["2.25 : Float"])
222
223     def test_integer_with_float_op_raises_type_error(self) -> None:
224         # +. requires Float operands; integers must not be accepted
225         with self.assertRaises(TypeError):
226             run_pipeline("x = 2 +. 3; print(x);")
227
228     def test_mixed_int_float_addition_raises_type_error(self) -> None:
229         # Integer + Float is no longer valid; must use +.
230         with self.assertRaises(TypeError):
231             run_pipeline("x = 1 + 2.5; print(x);")
232
233     def test_string_variable_type_inferred(self) -> None:
234         result = run_pipeline('x = "hello"; print(x);')
235         self.assertEqual(result.outputs, ['"hello" : String'])
236
237     def test_boolean_literal_type_inferred(self) -> None:
238         result = run_pipeline("x = true; print(x);")
239         self.assertEqual(result.outputs, ["true : Boolean"])
240
241     def test_boolean_equality_raises_type_error(self) -> None:
242         # Boolean == Boolean is not allowed; == only accepts arithmetic
243         operands
244         with self.assertRaises(TypeError):
245             run_pipeline('x = true; if (x == false) { print("then"); }'
246                           'else { print("else"); }')
247
248     def test_mixed_int_float_equality_raises_type_error(self) -> None:
249         # == and != must not accept mixed Integer/Float (no overloading
250         )
251         with self.assertRaises(TypeError):
252             run_pipeline("x = 5; y = 3.14; if (x == y) { print(x); }")
253
254     def test_mixed_int_float_inequality_raises_type_error(self) -> None:
255         :
256         with self.assertRaises(TypeError):
257             run_pipeline("x = 5; y = 3.14; if (x != y) { print(x); }")
258
259     def test_arithmetic_on_string_raises_type_error(self) -> None:
260         with self.assertRaises(TypeError):
261             run_pipeline('x = "a" + 1;')
262
263     def test_if_non_boolean_condition_raises_type_error(self) -> None:
264         with self.assertRaises(TypeError):
265             run_pipeline("if (1) { x = 2; }")
266
267     def test_while_non_boolean_condition_raises_type_error(self) ->
268         None:
269         with self.assertRaises(TypeError):
270             run_pipeline("x = 0; while (x) { x = x + 1; }")
271
272     def test_assignment_type_mismatch_raises_type_error(self) -> None:
273         with self.assertRaises(TypeError):

```

```

269         run_pipeline('x = 5; x = "hello";')
270
271     def test_undefined_variable_raises_type_error(self) -> None:
272         with self.assertRaises(TypeError):
273             run_pipeline("print(z);")
274
275     def test_function_wrong_arg_count_raises_type_error(self) -> None:
276         with self.assertRaises(TypeError):
277             run_pipeline("def f(a) { return a; } f(1, 2);")
278
279     def test_function_arg_type_mismatch_raises_type_error(self) -> None:
280         :
281         # first call infers a: Integer; second call with String must
282         # fail
283         with self.assertRaises(TypeError):
284             run_pipeline('def f(a) { return a; } f(1); f("hello");')
285
286     def test_uncalled_function_body_type_error_is_caught(self) -> None:
287         # type error inside body must be detected even if function is
288         # never called
289         with self.assertRaises(TypeError):
290             run_pipeline('def bad() { y = "hello" + 1; }')
291
292     def test_valid_uncalled_function_does_not_raise(self) -> None:
293         # a well-typed but uncalled function should not raise
294         result = run_pipeline("def add(a, b) { return a + b; }")
295         self.assertEqual(result.outputs, [])
296
297     #
298     -----
299
300     # Integration (full pipeline)
301     #
302     -----
303
304     class IntegrationTests(unittest.TestCase):
305         """End-to-end tests through all four pipeline stages."""
306
307         # --- Unary minus ---
308
309         def test_unary_minus_integer(self) -> None:
310             result = run_pipeline("x = -5; print(x);")
311             self.assertEqual(result.outputs, ["-5 : Integer"])
312
313         def test_unary_minus_float(self) -> None:
314             result = run_pipeline("y = -. 3.14; print(y);")
315             self.assertEqual(result.outputs, ["-3.14 : Float"])
316
317         def test_unary_minus_dot_variable(self) -> None:
318             result = run_pipeline("x = 1.5; y = -. x; print(y);")
319             self.assertEqual(result.outputs, ["-1.5 : Float"])

```

```

315
316 def test_unary_minus_variable(self) -> None:
317     result = run_pipeline("x = 10; y = -x; print(y);")
318     self.assertEqual(result.outputs, ["-10 : Integer"])
319
320 def test_unary_minus_inside_expression(self) -> None:
321     result = run_pipeline("x = 3 + -2; print(x);")
322     self.assertEqual(result.outputs, ["1 : Integer"])
323
324 # --- Control flow ---
325
326 def test_if_then_branch_executes(self) -> None:
327     result = run_pipeline(
328         'x = 5; if (x == 5) { print("yes"); } else { print("no"); }
329         ,
330     )
331     self.assertEqual(result.outputs, ['"yes" : String'])
332
333 def test_if_else_branch_executes_when_condition_false(self) -> None:
334     result = run_pipeline(
335         'x = 0; if (x == 5) { print("yes"); } else { print("no"); }
336         ,
337     )
338     self.assertEqual(result.outputs, ['"no" : String'])
339
340 def test_while_loop_executes_repeatedly(self) -> None:
341     result = run_pipeline("x = 3; while (x != 0) { print(x); x = x
342         - 1; }")
343     self.assertEqual(result.outputs, ["3 : Integer", "2 : Integer",
344         "1 : Integer"])
345
346 # --- Division by zero ---
347
348 def test_integer_division_by_zero_raises_error(self) -> None:
349     with self.assertRaises(InterpreterError):
350         run_pipeline("x = 10 / 0; print(x);")
351
352 def test_float_division_by_zero_raises_error(self) -> None:
353     with self.assertRaises(InterpreterError):
354         run_pipeline("x = 10.0 /. 0.0; print(x);")
355
356 # --- Functions ---
357
358 def test_function_integer_return(self) -> None:
359     result = run_pipeline("def square(n) { return n * n; } print(
360         square(7));")
361     self.assertEqual(result.outputs, ["49 : Integer"])
362
363 def test_function_type_inference(self) -> None:
364     result = run_pipeline(
365         "def add(a, b) { return a +. b; } result = add(2.0, 3.5);
366         print(result);"

```

```

361     )
362     self.assertEqual(result.outputs, ["5.5 : Float"])
363     self.assertIn("result          variable   Float", result.
364                   checked_scope.format_table())
365
366 def test_function_value_parameter_passing(self) -> None:
367     # Modifying a parameter inside a function must not affect the
368     caller's variable
369     result = run_pipeline(
370         "def inc(a) { a = a + 1; return a; } x = 10; y = inc(x);
371         print(x); print(y);"
372     )
373     self.assertEqual(result.outputs, ["10 : Integer", "11 : Integer
374                                     "])
375
376 def test_nested_function_calls(self) -> None:
377     result = run_pipeline(
378         "def add(a, b) { return a + b; } print(add(add(1, 2), 3));"
379     )
380     self.assertEqual(result.outputs, ["6 : Integer"])
381
382 # --- print() with all four types ---
383
384 def test_print_integer(self) -> None:
385     result = run_pipeline("print(42);")
386     self.assertEqual(result.outputs, ["42 : Integer"])
387
388 def test_print_float(self) -> None:
389     result = run_pipeline("print(3.14);")
390     self.assertEqual(result.outputs, ["3.14 : Float"])
391
392 def test_print_boolean(self) -> None:
393     result = run_pipeline("print(true);")
394     self.assertEqual(result.outputs, ["true : Boolean"])
395
396 def test_print_string(self) -> None:
397     result = run_pipeline('print("plc");')
398     self.assertEqual(result.outputs, ['"plc" : String'])
399
400 # --- Boolean expressions as values ---
401
402 def test_comparison_result_assigned_to_variable(self) -> None:
403     result = run_pipeline("x = 5; y = 5; flag = (x == y); print(
404         flag);")
405     self.assertEqual(result.outputs, ["true : Boolean"])
406
407 def test_comparison_result_printed_directly(self) -> None:
408     result = run_pipeline("x = 10; print(x != 0);")
409     self.assertEqual(result.outputs, ["true : Boolean"])
410
411 def test_comparison_result_false(self) -> None:
412     result = run_pipeline("a = 1; b = 2; eq = (a == b); print(eq);"
413     )

```

```

408         self.assertEqual(result.outputs, ["false : Boolean"])
409
410     # --- Reassignment ---
411
412     def test_variable_can_be_reassigned_same_type(self) -> None:
413         result = run_pipeline("x = 1; x = 2; x = 3; print(x);")
414         self.assertEqual(result.outputs, ["3 : Integer"])
415
416     def test_scope_isolation_if_block(self) -> None:
417         # Variable defined inside an if-block must not be visible
418         outside it
419         with self.assertRaises(TypeError):
420             run_pipeline(
421                 "x = 1; if (x == 1) { inner = 99; } print(inner);"
422             )
423
424     def test_string_equality_raises_type_error(self) -> None:
425         # == is only valid between arithmetic (Integer/Float) operands
426         with self.assertRaises(TypeError):
427             run_pipeline('a = "hello"; b = "world"; if (a == b) { print
428                 (a); }')
429
430     def test_integer_floor_division(self) -> None:
431         # 7 / 2 must be 3 (floor), not 3.5
432         result = run_pipeline("x = 7 / 2; print(x);")
433         self.assertEqual(result.outputs, ["3 : Integer"])
434
435     def test_void_function_assignment_raises_type_error(self) -> None:
436         # Assigning the result of a void function (no return) must be a
437         type error
438         with self.assertRaises(TypeError):
439             run_pipeline("def greet() { print(42); } x = greet();")
440
441     def test_builtin_print_assignment_raises_type_error(self) -> None:
442         with self.assertRaises(TypeError):
443             run_pipeline("x = print(42);")
444
445 if __name__ == "__main__":
446     unittest.main()

```

Listing 8.13: tests/test\_pipeline.py — automated regression suite (75 tests)